

Coalition Deception and Defensive Adaptation in a Social-Deduction Benchmark

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Abstract

Social-deduction games provide a compact setting for studying strategic communication under partial observability. We develop an Among Us-inspired benchmark with structured claims, exact truth labels, scripted and learned crews, and two impostors sharing a reinforcement-learning policy. An implementation audit identified incomplete-trajectory credit assignment, evaluation selection bias and unequal retention of private observations. We report completed components of a corrected replication separately from archived experiments awaiting replacement. Across ten seeds at each of three population sizes, meeting-only training increases held-out false ejections and aligned coalition voting. In corrected single-meeting spatial experiments, a learned defender has more false ejections than the scripted defender against the same frozen coalition. Coalition retraining increases this error at 5+2 and 7+2 players but decreases its average at 3+2; transfer to the scripted defender is limited. A separate 540,000-episode scripted mechanism study finds that credibility weighting performs worse than the mean against a coordinated alibi at the two larger sizes, despite lower average coalition credibility weight than its uniform share. Removing alibi support or moving caught-lying penalties changes errors across attack conditions and does not provide a uniformly beneficial repair. A stylized mean-field model separately characterizes corroboration-based weight amplification and its dependence-aware attenuation. Multi-round adaptation, adaptive defense comparisons and controls remain under replication; their archived measurements are provisional. The completed results are descriptive pending the full analysis and do not establish convergence or general robustness against adaptive coalitions.

1 Introduction

In a social-deduction game, agents make decisions from incomplete and conflicting evidence. Players report where they were and whom they saw, and a voting rule combines their reports into an ejection decision. Testimony is strategic: speakers choose what to say while pursuing potentially conflicting objectives. Agreement between two speakers can reflect independent observations or coordination within a hidden coalition.

We study this distinction in a controlled environment inspired by *Among Us*. Two impostors share a reinforcement-learning policy and face either scripted crews using fixed aggregation

rules or learned crews trained against a frozen coalition. The same engine supports scripted coalitions and coalitions trained against a particular defense. Our question is:

Does a defense’s performance against scripted coordinated testimony transfer to a coalition trained against that defense?

Status of the evidence in this working revision. An implementation audit found a rollout credit-assignment defect in variable-length spatial episodes, evaluation selection bias and unequal retention of private histories between scripted and learned policies. The synchronized, fixed-length meeting-only F1 collector was unaffected by that particular rollout defect, but its old evaluations reused training environment draws. Corrected F1 and F3 replications have now completed all ten seeds at all three population sizes; Sections 6.1 and 6.3 report their new held-out measurements. Section 6.2’s mechanism subsection reports a separate completed scripted replication (Tables 5 and 6). These results are descriptive; the full 240-job study and its planned inference are not yet complete. Remaining benchmark results are explicitly archived and provisional, including the F2 seven-rule comparison, multi-round learning, adaptive defense comparisons and controls. They predate the repaired collector or other implementation changes and cannot be combined with the new measurements. Appendix C specifies the corrected protocols. The theoretical calculations are independent of these implementation repairs.

The paper organizes the completed and pending evidence into eight findings and three controls.

1. **Learned coalition behavior (F1, corrected).** Across ten seeds and three crew sizes, independent untrained-to-final evaluations show increases in false ejections, same-target voting and creator survival. Direct false-claim changes are reported alongside these outcomes.
2. **Rules against scripted coalitions (F2, archived comparison).** In the meeting-only sweep the mean and credibility rules have no observed false ejections with truthful testimony. Against a coordinated alibi, soft and sharp credibility perform worse than the mean at 5+2 and 7+2. The median and trimmed mean discard useful sparse testimony.
3. **Hypothesis scoring (F2, archived comparison).** A crew that enumerates candidate coalitions reduces alibi false ejections from 0.329 to 0.156 at 5+2 and from 0.206 to 0.084 at 7+2. Its normalized archived scores used a simplified mixed-honesty model and smoothing; they were not exact posteriors for the game, and the rule has no demonstrated immunity to manufactured testimony.
4. **Staged adaptation (F3 corrected; F4 pending).** The corrected single-meeting runs retain higher false-ejection rates for the learned crew against C0. A coalition retrained against that crew increases false ejections at 5+2 and 7+2, but lowers the average at 3+2. The archived multi-round response pattern remains provisional pending its corrected replication.
5. **Repeated adaptation (F5, archived).** Over the observed generations, independently initialized response policies repeatedly change the outcome in favor of the side trained last. This is finite-horizon evidence, not a proof of non-convergence or an

attracting cycle.

6. **Analytical model (F6).** A deterministic mean-field surrogate gives conditions for corroboration-based weight amplification and its dependence-aware attenuation. Finite-sample simulations need not obey the surrogate’s thresholds exactly.
7. **Dependence-aware defense (F7, archived).** A penalty on concentrated agreement reduces the scripted alibi’s false-ejection rate in the spatial game, but the archived adaptive evaluations show little corresponding benefit against learned coalitions.
8. **Adaptive attack on hypothesis scoring (F8, archived).** Directly trained coalitions increase the hypothesis crew’s false-ejection rate from 0.079 to 0.359 at 5+2 and from 0.05 to 0.51 at 7+2. Cross-play and ballot diagnostics suggest that uninformative claims and abstention contribute; targeted interventions would be needed to identify their separate causal effects.

The controls remove the explicit private-symbol channel and partner-identity feature at the C0 stage, increase the defender’s training budget, and vary rewards. The tested C0 ablations retain strong coalition performance, but do not test an entire adaptation lineage or remove every route to role information. The budget and reward variants retain positive last-mover gains in the archived runs; they do not establish equivalence or exclude effects of training design.

We use **coalition** for agents sharing a policy and objective. The experiments measure their behavior in this benchmark, rather than the emergence of a coalition from independently incentivized agents or a universal ranking of voting rules.

2 Related work and contribution

Strategic information transmission was formalized by Crawford and Sobel’s model of cheap talk [1], where conflicting preferences produce coarse communication equilibria. Lewis’s signaling game provides the cooperative counterpart in which shared objectives support emergent conventions [2]. Reinforcement-learning studies such as EGG [3] and *Cheap Talking Algorithms* [4] bring these signaling problems into trainable sender-receiver systems. Work on multiple senders studies how coalition size and disagreement affect information aggregation [5], [6].

Related defenses include robust aggregation, Byzantine-resilient multi-agent reinforcement learning, and peer-prediction mechanisms [6], [7], [8]. Their guarantees have explicit assumptions; arbitrary Byzantine behavior is not equivalent to a fixed scripted attack. Results on the geometric median also identify limits to strategyproofness [9]. Social-deduction systems such as *Hidden Agenda* and language-model studies provide a second line of work on learned deception, hidden roles, and strategic discussion [10], [11]. Work on adversarial communication, multi-agent credit assignment, emergent cooperation under uncertain incentives, and secret collusion provides adjacent models [12], [13], [14], [15].

Evaluating a defense with attacks adapted to it is an established requirement in adversarial robustness [16]. Learning approximate responses to policies or policy populations is also established in multi-agent reinforcement learning [17]. Our contribution is an empirical com-

parison within a structured social-deduction benchmark, rather than the general observation that a defense needs adaptive evaluation.

The comparison uses common defenses within each environment configuration and contrasts scripted with trained adversaries. The meeting-only and spatial games differ, so comparisons across those configurations are not controlled changes of attack policy alone. Four further elements make the comparison possible and interpretable.

1. **An interpretable environment.** Private evidence, public claims, response actions, votes, roles, and state transitions are recorded explicitly. Structured claims allow truth and support labels to be computed without interpreting natural-language dialogue, and the same meeting engine accepts scripted coalitions with a fixed dependence structure and learned coalitions that choose their own.
2. **A mechanism-level comparison of aggregators.** The engine can use order-statistic rules, credibility weighting, a dependence-aware discount, or a heuristic hypothesis-scoring rule, and every rule’s realized weights can be measured. This is what makes it possible to say *through which channel* a scripted attack acts, and therefore whether a learned coalition uses the same channel.
3. **A staged and a long-horizon adaptation study.** The $C0 \rightarrow D1 \rightarrow C1$ protocol separates initial coalition learning, defender response, and coalition adaptation; its continuation for ten generations, with cross-generation play and policy-distance tracking, describes the finite-horizon sequence of approximate responses trained from fresh initializations. Controls for the coalition’s coordination channel, the training budget, and the reward specification test whether the dynamics are properties of the interaction.
4. **An analytical model** of credibility weighting under correlated testimony, with exact agreement probabilities and a deterministic surrogate checked alongside Monte Carlo simulations and the game’s measured credibility weights, so that the one failure mode the model describes, corroboration capture, can be distinguished from the ones the benchmark exhibits.

3 Environment

Sections 3–5 describe the benchmark and identify differences between the archived and corrected protocols. Appendix C records the corrected replication. F1, F3 and the scripted mechanism diagnostic now use that corrected implementation; the remaining empirical sections identify their archived results explicitly.

3.1 Game structure

The benchmark contains two hidden impostors and a variable number of crew members. The main crew sizes are 3, 5, and 7, producing 3+2, 5+2, and 7+2 players. Every experiment uses one of two configurations of the same engine (Table 1).

The meeting-only game generates evidence exogenously and then runs a structured meeting, which makes the scripted rule comparison of F2 easy to reproduce with labeled evidence.

Table 1. The two configurations. Both use the same claim, response, vote, truth-labeling and transition code; roles are re-randomized every episode, the reporter announces the incident time, and roles are not revealed on ejection.

	Meeting-only game (F1, F2)	Spatial short game (F3 to F5, F7, F8, controls)
Rooms	Hub, Electrical, Navigation, Medbay	same
Edges	Hub–Electrical, Hub–Navigation, Hub–Medbay	the same three plus Electrical–Navigation
Evidence	positions sampled by a random walk on the map (stay probability 0.35), 6 free-play turns	positions produced by the agents’ movement actions, 8 free-play turns, two tasks per agent
Incident	one incident placed by the episode process	created by an impostor action with one crew target and no other living crew in the room
Coalition channel	none	learned private symbol channel, four symbols
Meeting	claim round, response round, vote (plurality, ties do not eject)	same
Rounds	one meeting	one meeting (F3) or up to eight rounds with terminal win conditions (F4 onward)

The spatial short game produces evidence through movement, tasks, and an isolation-gated incident action, and adds a learned low-bandwidth private channel between the two impostors. The spatial map is the three-edge star of the meeting-only game plus one extra edge between Electrical and Navigation, so a player can move between those rooms without passing through the Hub.

3.2 Information and actions

Each player receives a private view containing its own evidence and a public transcript containing the claims and responses already made. Roles remain hidden in the public transcript. Actions are factorized into legal heads: movement between connected rooms; a task, observe, or report action; an incident action for an impostor when the local conditions permit it; a structured claim about a room, time, or observed player, with a polarity; a response to another claim (accuse, defend, agree, contradict, or neutral); a private symbol in the spatial game; and a vote for a player or skip.

Claims are structured symbolic objects. Claims have a truth/support label (true, unsupported, false, or contradictory) relative to the recorded state and the speaker’s evidence, which provides exact measurement of strategic testimony. No-information is a separate claim action, not a fifth truth label. The scripted policies reconstruct the observer’s full within-round co-location history, including negative observations, to check claims. In the archived experiments, learned actors instead received the most recent sighting and elapsed time for each agent, their current room, and the public transcript. Their non-recurrent observation did not preserve the full private trajectory, confounding comparisons between scripted and learned defenders. The corrected spatial replication adds a time-indexed private-history block that retains the same positive and negative positional evidence without exposing unseen rooms or future positions.

The two impostors know one another and share a policy. An identity feature marks the partner in each impostor’s observation so that the two copies of the shared policy can specialize within the coalition. In the spatial game the coalition also has the learned symbol channel. Both the identity feature and the channel are ablated in Section 6.5.

3.3 Scripted policies

The scripted **truthful crew** moves toward its task rooms, reports an incident when it finds the marker, and in the meeting states its most informative *supported* claim: among the claims its own trajectory entitles it to make, it prefers claims about other agents over claims about itself, positive sightings over negative ones, and claims at or adjacent to the incident time and in the incident room. It speaks the claim’s own polarity, so a crew member that did not see an agent in its room says so rather than asserting the opposite. It accuses the most suspicious agent when its suspicion exceeds a threshold and votes the same way, skipping otherwise. Its suspicion is formed by the aggregation rule of Section 3.4.

Four scripted **coalitions** define F2. The *truthful* coalition uses supported claims and the crew’s suspicion rule, while avoiding its partner in voting. For the *lone liar*, the creator

claims to have been elsewhere, accuses a selected innocent and votes for that target; the partner speaks truthfully without defending the creator. In the *coordinated alibi*, the creator makes a self-location claim away from the scene, while the partner claims to have seen the creator away from it. The creator accuses an innocent, the partner defends the creator, and both vote for the selected innocent. Without a channel, their asserted rooms are not guaranteed to agree. The *coordinated framer* has both members claim to have seen the selected innocent in the incident room at the incident time, then accuse and vote for that player. Those framing claims are not necessarily truthful. These are distinct scripts, not pure manipulations of one agreement variable.

In the archived spatial game, scripted coalitions inspected true occupancy in adjacent rooms when choosing movement. That privileged heuristic was unavailable to learned actors. The corrected scripts use their own last sightings, public alive status, and legitimately known partner and marker information. F2’s exogenous evidence removes this particular movement difference.

3.4 Aggregation rules

A crew voter forms a suspicion score over the other agents from two sources, its own grounded evidence and testimony, and the aggregation rule decides how testimony is combined:

$$\text{suspicion}_v(j) = \text{own}_v(j) + \text{AGG}_s [\text{signal}_v(s, j)],$$

where $\text{signal}_v(s, j)$ is how much speaker s ’s claims and responses implicate candidate j as judged by voter v . A claim that places j in the incident room at the incident time raises j ’s suspicion; a claim by another speaker that places j elsewhere lowers it (the *alibi-witness* term); an accusation raises it; and a speaker whose claim the voter personally knows to be false is implicated only itself, by a large *caught-lying* penalty that sits in the liar’s own row. The voter’s own evidence marks agents it saw at the scene, clears agents it saw elsewhere, and treats agents it cannot place as suspects by default.

- **Mean:** the sum of the per-speaker rows.
- **Median and trimmed mean:** the per-candidate median, or the mean after discarding the extreme row on each side, rescaled to the sum’s scale.
- **Soft and sharp credibility:** a weighted sum in which a speaker’s weight is the softmax of its credibility: -3 if the voter catches a false claim, otherwise -2 for a self-contradictory claim set, otherwise the count of other informative speakers with jointly satisfiable claims; the temperatures are 1.5 (soft) and 0.15 (sharp).
- **Dependence-aware credibility:** soft credibility whose weights are multiplied by $e^{-\lambda \text{ex}_s}$, where ex_s is how much more s agrees with its closest partner than with everyone on average. Agreement is pooled from the public record over four channels (joint satisfiability of claims, one speaker clearing the other of the scene, defend or agree responses, and the same non-skip vote) over a sliding window of three meetings; $\lambda = 4$. A **dependence-weighted vote** applies the same discount to ballots instead of to

testimony. This ballot-level rule remains active when a learned crew replaces the scripted testimony aggregator.

- **Archived hypothesis elimination (hypothesis scoring):** this version enumerated size-two coalitions among all agents other than the voter. For each candidate it assumes that speakers outside the coalition are truthful and averages over branches in which each coalition speaker’s informative claims are treated as truthful with probability $1/2$ or ignored otherwise. Caught lies and joint inconsistencies invalidate branches. A candidate creator receives a location factor of R when placed at the scene, 1 when unplaced, and 0 when placed elsewhere. The implementation sums these factors, assigns weight 10^{-3} to zero-weight hypotheses, normalizes, and votes for the leading living candidate only if its membership score exceeds $1/2$. Creator score and own evidence provide small tie-breaking adjustments. The rule ignores response accusations. Coalition claims are retained in truthful branches even under the true coalition hypothesis; consequently the procedure has no general invariance to manufactured corroboration. Its uniform-room factors, incomplete spatial constraints and smoothing make it a likelihood-inspired heuristic, rather than an exact posterior for the benchmark. The candidate set also includes incident victims, even though victims are necessarily crew; these archived baseline results require comparison with a corrected model.

The corrected hypothesis rule excludes publicly known crew, including murdered victims inferred from public death and ejection records. Accepted claim constraints and private positive/negative observations are solved jointly, with public inactivity respected. It averages feasibility indicators over the declared honesty gates and auxiliary creator/time assignments, normalizes positive scores, and gives infeasible candidates zero weight; an all-zero view skips when legal. Membership, creator score, own evidence, and agent index provide exact lexicographic tie breaking. This remains a feasibility heuristic, not a calibrated posterior for the full game-generating process. The shared solver now checks all emitted constraints using propagation and backtracking; some negative transition claims still use a documented necessary-constraint relaxation.

The credibility rules use exactly the same information as the mean; they differ only in how much each speaker’s row counts.

4 Learning procedure

4.1 PPO policy

The learned agents use a masked actor-critic policy trained with Proximal Policy Optimization (PPO) [18]. The actor maps an observation to a probability distribution over each action head; legal-action masks remove actions unavailable to that observer [19]. In the corrected hidden-partner intervention, incident masks allow role-blind attempts that can resolve as a legal no-op under hidden game rules, so mask entries do not reveal the partner’s role. The critic maps the observation to a scalar value estimate. The clipped actor objective is

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right],$$

where $\hat{A}_t$ is the estimated advantage and $r_t(\theta)$ the new-to-old action probability ratio. Both actor and critic are two-layer networks with 128 hidden units; the learning rate is 3×10^{-4} , $\gamma = 1$ for terminal episodic rewards, generalized advantage estimation uses $\lambda = 0.95$, and each update runs up to eight PPO epochs over four minibatches with an entropy coefficient annealed from 0.02 to 0.005. PPO can stop an update early when approximate KL exceeds 0.075 (1.5 times the target KL of 0.05). The policy is non-recurrent: the meeting observation contains the public transcript, so the visible history is available at each meeting, but long-horizon memory across rounds is a limitation of the multi-round study. The standard stage budget is 400 updates; defender-budget controls use 800 or 1,600. F1 collects 64 total episodes per update across 64 parallel environments. Spatial runs used 32 environments and a target of at least 32 completed episodes per update; asynchronous completion could overshoot this count. The original collector also finalized unfinished fragments at this boundary with zero bootstrap. The corrected collector waits for complete vector batches; all reported spatial training measurements predate that fix. Evaluation samples actions from the policy, as during training.

4.2 Reward design

The reward is terminal. Directly rewarding “lie more”, “agree with your partner”, or “protect the killer” would program the behavior that the experiment is supposed to measure.

In F1 the coalition uses the incident-gated **false_ejection** objective: an innocent ejection during a real incident gives the coalition +1, while a skip, a coalition ejection, or a meeting without an incident gives −1. Only the coalition is trained; the crew follows the scripted soft-credibility policy. F2 is fully scripted.

In the single-meeting response cycle (F3) and its budget control every stage uses the **balanced_ejection** objective, which is zero-sum: the coalition receives +1 for an innocent ejection, 0 for a skip, and −1 for a coalition ejection or for a meeting without an incident, and the crew receives the negative of the coalition’s score. The trained side optimizes its own score while the opposing checkpoint is fixed. The single-meeting multi-generation loop (F5 and F7) keeps **balanced_ejection** for the coalition stages and trains the crew stages on **meeting_ejection**, under which only a real incident followed by an ejection is scored (a coalition ejection is +1 for the crew, an innocent ejection −1, skips and incident-free meetings 0): the defender is trained on the quality of real meeting decisions, while the coalition is also penalized for failing to create an incident. The two constructions are compared directly by the reward control of Section 4.4.

For the multi-round game (F4 onward) the objective is terminal survival: the coalition wins when, after a meeting vote, the number of living impostors is at least the number of living crew; the crew wins when the coalition is eliminated or the round cap is reached without coalition parity. The reward control also replaces survival by a framing-count objective

(`false_ejection`: the coalition’s terminal reward counts innocents ejected in rounds with a real incident; the crew’s is its negative).

4.3 Staged and multi-generation training

The response-cycle experiments use frozen checkpoints:

```
C0: coalition learns against the scripted soft-credibility crew
  ↓ freeze C0
D1: crew learns against frozen C0
  ↓ freeze D1
C1: coalition learns against frozen D1
```

This order asks how a defender policy responds to a known learned coalition and how much of that response survives when the coalition is trained again. It is a sequential response process; simultaneous co-evolution is a separate condition.

Each new stage initializes a fresh actor and critic; learner weights are not warm-started from the preceding same-side policy. The frozen opponent is loaded from its checkpoint. Thus policy distances include effects of independent training and initialization.

The cycle is continued as $C0 \rightarrow D1 \rightarrow C1 \rightarrow D2 \rightarrow C2 \rightarrow \dots \rightarrow Dk \rightarrow Ck$. Every stage trains one side for the same budget against the other side’s frozen checkpoint; D0 is the scripted soft-credibility crew, and stage s of a run with seed σ uses training seed $\sigma + 10000s$, so the first three stages use the same seeds and budget as the F3 and F4 runs. After every stage the frozen match-up is evaluated on 1,000 episodes, both sides’ policies are compared with their predecessors on a fixed scripted probe set whose size depends on the configuration (10,920 decision points in the 5+2 multi-round game) (Kullback–Leibler and Jensen–Shannon divergences summed over masked action heads and averaged over probe rows, a behavioral Jensen–Shannon divergence between vote distributions in canonical categories, and the fraction of greedy actions that differ), and after the last stage every C_i is played against every D_j (500 episodes per cell). Ten training seeds are run for $k = 10$ at 5+2 in both regimes and at 3+2 in the multi-round game; five seeds are run for $k = 3$ at 7+2 in the multi-round game.

4.4 Controls

Coordination ablation. The C0 stage of the spatial game (5+2 players, single meeting and multi-round) is trained under four information conditions: the default (partner identity known, private channel on), no channel, partner identity hidden, and neither. Five seeds each.

Defender budget. The single-meeting cycle at 5+2 is repeated with the defender stage trained for 800 and 1,600 updates instead of 400, and the multi-round cycle with 800, while C0 and C1 keep 400 updates; five seeds, the same seeds as the reference runs.

Reward specification. The single-meeting loop at 5+2 is repeated for four generations with one objective for both sides, either `balanced_ejection` (the F3 construction) or

`meeting_ejection`, so that either the crew’s reward for incident-free meetings is introduced or the coalition’s incident-free penalty is removed; both objectives give zero reward for a skip following an incident; the multi-round loop at 5+2 is repeated for two generations with the framing-count objective in place of survival. Five seeds each, the same seeds as the reference loops, so every comparison is paired by seed.

5 Evaluation protocol

The primary single-meeting outcome is the **false-ejection rate**, the probability that a crew member is ejected, because it directly measures the central harm to the crew. It is reported together with the **incident-creator ejection rate**, since a rule that skips whenever it is unsure has a low false-ejection rate for the wrong reason. In the spatial game the coalition decides whether an incident occurs and a meeting is held either way, so single-meeting outcomes there are reported per meeting together with the **incident rate** and the incident-conditional rates (Section 6.3); the creator-ejection rate always requires a real incident. Secondary outcomes are the coalition-favorable rate (no coalition member ejected), the coalition game-win rate in multi-round play, same-target voting between the two impostors, the coalition’s false-claim rate (the fraction with the FALSE label; contradictory and no-information claims are separate categories), creator survival, and rounds played. For multi-round games, the legacy FE, claim and same-target summaries describe the terminal meeting; game-win rate and rounds played describe the whole game. Section 6.4 adds a separately recomputed cumulative incident-bearing harm measure. F1 originally evaluated with the training seed and episode indices. F2 uses condition/rule-specific evaluation streams; spatial cross-play generally uses 987654321+ run seed, separate from training. The corrected Trainer default also uses this offset unless an evaluation seed is supplied explicitly.

The main analyses use: F1, learned-coalition training curves at crew sizes 3, 5, and 7 with ten seeds each; F2, seven rules against four scripted coalitions at three crew sizes, ten seeds and 1,000 episodes per cell; F3 and F4, ten seeds per crew size in the response cycle with 1,000 cross-play episodes per stage; F5, the multi-generation loop with ten seeds at 5+2 in the multi-round game and in the single meeting, ten at 3+2 and five at 7+2; F6, the analytical surrogate compared with Monte Carlo (4,000 trials per grid point); F7, the dependence-aware defense against scripted and frozen learned coalitions (1,000 episodes per cell, five seeds), against freshly retrained coalitions (ten seeds at 5+2 and five at 3+2), and inside the multi-generation loop (ten seeds); F8, a fresh coalition trained against the hypothesis-elimination crew and cross-evaluated against the three selected rules (ten seeds at 5+2, five at 7+2); and the three controls with five seeds each.

The run seed is the independent statistical unit; in fully scripted experiments it is an evaluation replicate rather than a training seed. Evaluation games provide within-seed precision and are not treated as independent training replicates. Paired exact sign-flip tests for the archived results are reported as descriptive consistency checks; with ten nonzero paired differences the smallest attainable two-sided value is $2/2^{10} = 0.001953125$ (reported as 0.002), and with five it is $2/2^5 = 0.0625$ (reported as 0.063). Tests enumerate signs of the paired mean difference and assume sign symmetry under the null; they are unadjusted exploratory

comparisons, not independent confirmatory tests. A non-significant contrast does not establish equivalence. Conditional rates are first computed within each seed and then averaged, so aggregate conditional and unconditional means need not satisfy a pooled mixture identity; zero-denominator conventions are noted with the affected tables.

The corrected replication uses independent monitored and final evaluation streams, fixed episode indices, and isolated policy-sampling generators. Its dated analysis plan pairs seed IDs and specifies effect sizes, paired bootstrap intervals, and Holm correction within fixed comparison families. No new inferential results are reported until the required runs and comparisons are complete. These procedures do not retroactively change the archived exploratory p-values.

6 Results

6.1 F1: learning against a fixed scripted defense

The repaired implementation provides a genuine untrained baseline and a separate holdout comparison after 400 PPO updates. All 30 F1 jobs are complete. In the independent holdout, false-ejection rates increase from 0.045 to 0.271 at 3+2, from 0.073 to 0.212 at 5+2, and from 0.091 to 0.194 at 7+2 (Figure 1, Table 2). These replace the archived comparison whose first recorded endpoint already followed 20 updates.

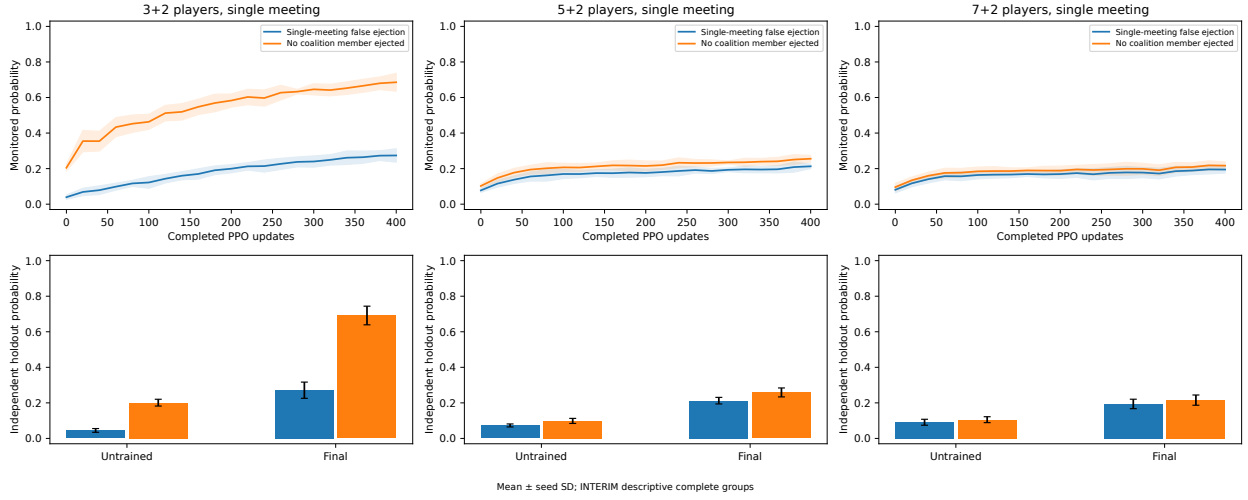


Figure 1. Corrected meeting-only replication against the scripted soft-credibility crew, with all ten training seeds at each of three population sizes. Top: monitored curves at completed PPO updates, using a separate evaluation stream. Bottom: untrained and 400-update checkpoints on the independent final evidence stream. Each evaluation uses 400 episodes per seed; lines, shading and bars show seed means and sample SDs. “No coalition member ejected” includes skips and is not a game-win probability. These complete-group results are descriptive while the remaining study is pending.

Vote coordination and creator survival also rise at all three population sizes. In contrast to the archived endpoint account, the mean explicitly false-claim fraction falls at every population: 0.429 to 0.328, 0.423 to 0.349, and 0.420 to 0.385. Thus increased harm accompanies more coordinated voting and fewer explicitly false claims in these runs. These behavioral co-occurrences do not establish that reducing false claims causes the increase in harm, nor do they identify a unique mechanism involving omissions, compatible testimony, or voting.

Figure 1 also separates false ejection from the broader event that no coalition member is ejected. That event includes a skip, so an improvement in coalition survival need not imply an increase of the same size in innocent-player harm. The table’s distinct outcome definitions are needed to interpret what the policy has learned.

Table 2. Corrected untrained and final holdout outcomes, mean $\pm$ sample SD across all ten training seeds per population, with 400 evaluation episodes per endpoint and seed. Final means exactly 400 PPO updates. Same-target voting requires both coalition votes to name the same agent; shared skips do not count. Creator survival is the per-episode probability that a real incident occurred and its creator was not ejected. The last column is the fraction of coalition claims labeled false, not the probability that an episode contains a false claim. SD describes variation across seeds; no confidence intervals or significance tests are reported during the interim analysis.

Population, checkpoint	False ejection	Same-target voting	Creator survival	Coalition false claims
3+2, untrained	0.045 $\pm$ 0.011	0.126 $\pm$ 0.023	0.457 $\pm$ 0.024	0.429 $\pm$ 0.017
3+2, final	0.271 $\pm$ 0.046	0.613 $\pm$ 0.062	0.789 $\pm$ 0.036	0.328 $\pm$ 0.074
5+2, untrained	0.073 $\pm$ 0.008	0.108 $\pm$ 0.023	0.310 $\pm$ 0.020	0.423 $\pm$ 0.012
5+2, final	0.212 $\pm$ 0.018	0.316 $\pm$ 0.058	0.438 $\pm$ 0.023	0.349 $\pm$ 0.046
7+2, untrained	0.091 $\pm$ 0.017	0.092 $\pm$ 0.016	0.272 $\pm$ 0.018	0.420 $\pm$ 0.011
7+2, final	0.194 $\pm$ 0.027	0.244 $\pm$ 0.041	0.366 $\pm$ 0.022	0.385 $\pm$ 0.048

6.2 F2: testimony rules against scripted coalitions

The honest baseline and the attacked conditions tell different stories (Figure 2, Tables 3 and 4). With honest testimony the mean, soft credibility, sharp credibility and the dependence-aware rule never eject an innocent at any crew size and eject the incident creator in nine meetings out of ten at 5+2 and 7+2. The median and the trimmed mean are much worse with the *same* honest testimony (false ejection 0.39 and 0.31 at 5+2): most speakers say nothing about most candidates, so the per-candidate median of the testimony rows is zero and the decision falls back on the voter’s own, mostly uninformative, evidence. Order statistics are the wrong tool for sparse testimony.

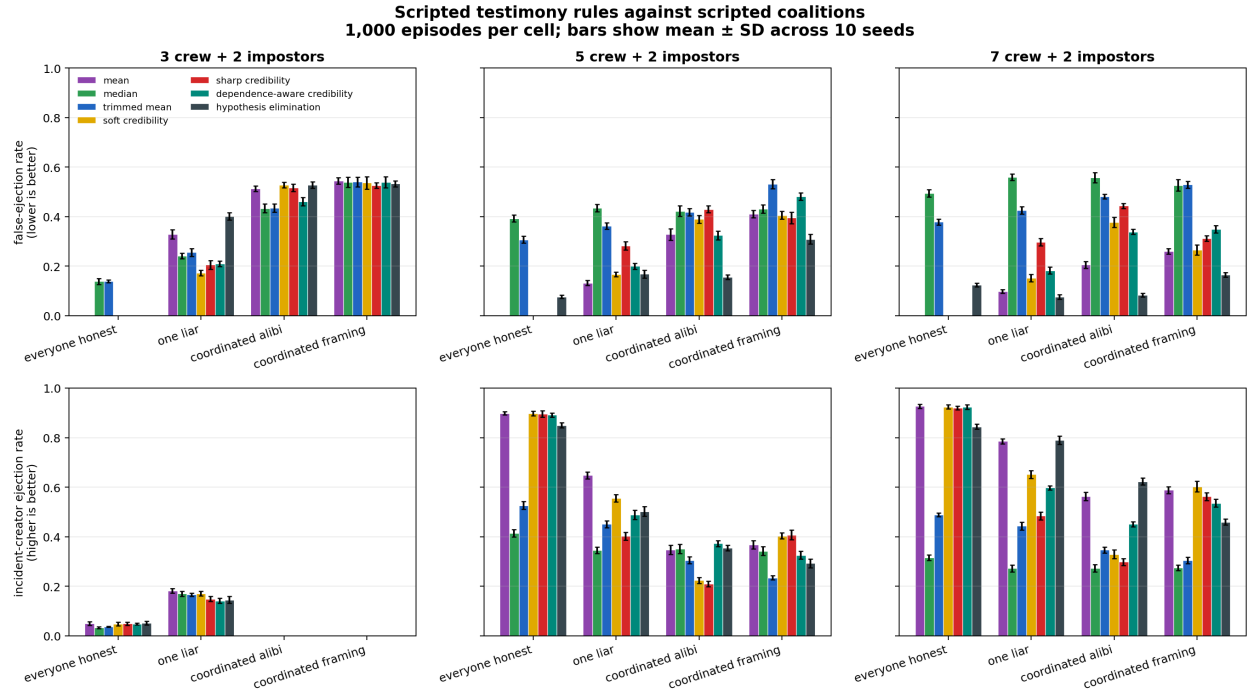


Figure 2. Archived measurements. False-ejection rate (top) and incident-creator ejection rate (bottom) of seven testimony rules against four scripted coalitions at three crew sizes. Bars show the mean $\pm$ SD across ten seeds; 1,000 episodes per cell.

Table 3. Archived measurements. False-ejection rate, mean across ten seeds (across-seed SDs are zero in observed zero-error cells and typically 0.01 to 0.03 otherwise). Trim. denotes trimmed mean and Hyp. hypothesis scoring; bold marks the best rule where the hypothesis crew leads.

Crew, condition	Mean	Median	Trim.	Soft cred.	Sharp cred.	Dep.-aware	Hyp.
3+2, honest	0.000	0.139	0.139	0.000	0.000	0.000	0.000
3+2, one liar	0.329	0.241	0.256	0.174	0.206	0.211	0.401
3+2, coordinated alibi	0.513	0.434	0.436	0.528	0.517	0.461	0.528
3+2, coordinated framer	0.546	0.540	0.541	0.537	0.527	0.540	0.533
5+2, honest	0.000	0.393	0.308	0.000	0.000	0.000	0.077
5+2, one liar	0.133	0.435	0.363	0.167	0.282	0.200	0.169
5+2, coordinated alibi	0.329	0.423	0.418	0.390	0.430	0.325	0.156
5+2, coordinated framer	0.412	0.432	0.532	0.406	0.396	0.481	0.310
7+2, honest	0.000	0.495	0.379	0.000	0.000	0.000	0.125
7+2, one liar	0.098	0.560	0.426	0.153	0.297	0.183	0.077
7+2, coordinated alibi	0.206	0.558	0.482	0.377	0.444	0.339	0.084
7+2, coordinated framer	0.260	0.527	0.529	0.266	0.312	0.349	0.166

Table 4. Archived measurements. Incident-creator ejection rate for the same cells (abbreviations as in Table 3) (at 3+2 the creator is essentially never ejected under the two coordinated attacks, because after the incident two coalition votes tie two crew votes and ties do not eject).

Crew, condition	Mean	Median	Trim.	Soft cred.	Sharp cred.	Dep.-aware	Hyp.
5+2, honest	0.90	0.41	0.53	0.90	0.90	0.89	0.85
5+2, one liar	0.65	0.35	0.45	0.56	0.40	0.49	0.50
5+2, coordinated alibi	0.35	0.35	0.31	0.22	0.21	0.37	0.36
5+2, coordinated framer	0.37	0.34	0.23	0.40	0.41	0.33	0.29
7+2, honest	0.93	0.32	0.49	0.93	0.92	0.92	0.84
7+2, one liar	0.79	0.27	0.44	0.65	0.48	0.60	0.79
7+2, coordinated alibi	0.56	0.27	0.35	0.33	0.30	0.45	0.62
7+2, coordinated framer	0.59	0.28	0.30	0.60	0.56	0.53	0.46

The coordinated alibi is where the rules separate. At 5+2 and 7+2 the mean outperforms soft credibility, and sharp credibility is worse still (false ejection 0.21, 0.38 and 0.44 at 7+2; the paired differences from the mean are positive in every seed, exact sign-flip $p = 0.002$). The credibility rules also eject the creator less often. Sophistication in the weighting therefore costs robustness against the coordinated alibi, and it costs more the sharper the weighting; the diagnostic below shows that the cost is not paid through the corroboration weights. The one-liar condition is intermediate: soft credibility is a little worse than the mean and sharp credibility clearly worse, while at 3+2, where one caught lie decides the meeting, both credibility rules beat the mean. The framer combines scene-placement claims with accusations and aligned votes. Its performance therefore cannot be attributed to accusations alone, and the ground-truth consistency of its claims can distinguish it from honest co-witnesses. The 3+2 setting is the parity-boundary case in which no rule does well against a coordinated attack.

6.2.1 Where the credibility rule’s behavior comes from

We conducted a separate mechanism replication using the repaired frozen implementation: ten evaluation replicates with 500 meetings each for every combination of three initial crew sizes, four scripted conditions, three aggregation rules, and three variants (540,000 meetings). Tables 5 and 6 report means and sample SDs across replicates. These new measurements supersede the former single-seed mechanism diagnostic; they are separate from the archived full F2 rule sweep above. Within each size, condition, and replicate, all nine rule/variant cells share the same exogenous evidence and episode identities. The interventions govern crew responses and votes, so subsequent public responses may differ; they do not hold the full transcript fixed.

“Coalition caught” is the fraction of honest voter decisions that privately refute at least one

Table 5. New pre-ejection soft-credibility diagnostic (mean $\pm$ SD across ten replicates of 500 meetings each). C/H denote coalition/honest speakers. Coalition weight is total normalized speaker weight; parentheses give its uniform share. All honest voters are included, including a voter subsequently ejected.

Game, condition	Credibility C	Credibility H	Coalition weight (uniform)	Coalition caught	Top pick differs
3+2, Truthful	3.000 $\pm$ 0.000	3.000 $\pm$ 0.000	0.5000 $\pm$ 0.0000 (0.5000)	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
3+2, Lone liar	1.581 $\pm$ 0.037	2.982 $\pm$ 0.006	0.4225 $\pm$ 0.0022 (0.5000)	0.4649 $\pm$ 0.0129	0.1264 $\pm$ 0.0076
3+2, Alibi	0.164 $\pm$ 0.075	2.960 $\pm$ 0.008	0.2630 $\pm$ 0.0059 (0.5000)	0.5583 $\pm$ 0.0147	0.1674 $\pm$ 0.0161
3+2, Framer	-1.547 $\pm$ 0.052	2.593 $\pm$ 0.040	0.1479 $\pm$ 0.0041 (0.5000)	0.7787 $\pm$ 0.0080	0.0029 $\pm$ 0.0014
5+2, Truthful	5.000 $\pm$ 0.000	5.000 $\pm$ 0.000	0.3333 $\pm$ 0.0000 (0.3333)	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
5+2, Lone liar	2.836 $\pm$ 0.054	4.987 $\pm$ 0.003	0.2606 $\pm$ 0.0017 (0.3333)	0.5353 $\pm$ 0.0138	0.0319 $\pm$ 0.0054
5+2, Alibi	0.778 $\pm$ 0.078	4.970 $\pm$ 0.004	0.1425 $\pm$ 0.0030 (0.3333)	0.6317 $\pm$ 0.0093	0.0769 $\pm$ 0.0085
5+2, Framer	-0.335 $\pm$ 0.081	4.476 $\pm$ 0.025	0.1062 $\pm$ 0.0035 (0.3333)	0.6921 $\pm$ 0.0125	0.0085 $\pm$ 0.0021
7+2, Truthful	7.000 $\pm$ 0.000	7.000 $\pm$ 0.000	0.2500 $\pm$ 0.0000 (0.2500)	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
7+2, Lone liar	4.068 $\pm$ 0.036	6.989 $\pm$ 0.002	0.1863 $\pm$ 0.0010 (0.2500)	0.5819 $\pm$ 0.0064	0.0174 $\pm$ 0.0036
7+2, Alibi	1.520 $\pm$ 0.124	6.973 $\pm$ 0.003	0.0976 $\pm$ 0.0027 (0.2500)	0.6535 $\pm$ 0.0104	0.0672 $\pm$ 0.0069
7+2, Framer	0.361 $\pm$ 0.057	6.402 $\pm$ 0.013	0.0630 $\pm$ 0.0018 (0.2500)	0.6830 $\pm$ 0.0080	0.0086 $\pm$ 0.0032

coalition claim. “Top pick differs” compares mean and soft credibility on the same soft-baseline record, restricted to eligible candidates. Credibility scores include refutation and self-contradiction penalties as well as pairwise compatibility. One crew member is removed by the incident before testimony: initial 3+2, 5+2, and 7+2 games have 2+2, 4+2, and 6+2 reporters. Appendix A counts actual reporters, so 3+2 is outside its strict-minority condition $m < n$.

Under truthful testimony, all speakers receive the same credibility score and the normalized weights are uniform; no same-record mean-versus-soft top-pick differences were observed. All truthful cells have zero observed false ejections, but this does not imply reliable identification at every size. In truthful 3+2 baseline meetings, the no-ejection rate is 0.9496 ± 0.0094 and the creator-ejection rate is 0.0504 ± 0.0094 . At truthful 5+2 and 7+2 every meeting ejects a coalition member, and creator-ejection rates are 0.9030 ± 0.0116 and 0.9204 ± 0.0113 ; remaining ejections select the partner.

For every attacked condition, the coalition’s average normalized weight is below its uniform share. In the coordinated alibi, the averages are 0.2630, 0.1425, and 0.0976 at initial 3+2, 5+2, and 7+2. Below-uniform averages alone do not exclude individual episodes of weight capture. A further audit of all 240,000 retained honest voter records found no coalition weight above its uniform share at numerical tolerance 10^{-12} . This is a descriptive result for these scripted records, not a general immunity claim. All speakers supplied informative claims in this experiment.

We tested two variants of each aggregation rule. “Alibi off” removes testimony-based alibi credit, including the positive WAS_WITH subject credit controlled by the same coefficient. “Catch outside” places the caught-lying suspicion penalty in the voter’s own evidence instead of within the speaker’s aggregated row. In the baseline credibility implementation, down-weighting a caught liar also reduces the self-incriminating penalty contained in that row.

Table 6. New false-ejection rates across all scripted conditions, rules and variants (mean $\pm$ SD across ten replicates of 500 meetings per cell). Soft and sharp denote credibility rules. Evidence is matched within each size/condition/replicate; responses and votes are regenerated. Differences are descriptive.

Game, condition	Rule	Baseline	Alibi off	Catch outside
3+2, Truthful	Mean	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
3+2, Truthful	Soft	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
3+2, Truthful	Sharp	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
3+2, Lone liar	Mean	0.3290 ± 0.0179	0.3662 ± 0.0184	0.3290 ± 0.0179
3+2, Lone liar	Soft	0.1734 ± 0.0205	0.2106 ± 0.0237	0.3274 ± 0.0178
3+2, Lone liar	Sharp	0.2010 ± 0.0230	0.2188 ± 0.0277	0.3230 ± 0.0175
3+2, Alibi	Mean	0.5176 ± 0.0249	0.4030 ± 0.0219	0.5180 ± 0.0250
3+2, Alibi	Soft	0.5288 ± 0.0238	0.3728 ± 0.0193	0.5046 ± 0.0223
3+2, Alibi	Sharp	0.5144 ± 0.0252	0.4406 ± 0.0222	0.4528 ± 0.0201
3+2, Framer	Mean	0.5354 ± 0.0195	0.5370 ± 0.0200	0.5354 ± 0.0195

Game, condition	Rule	Baseline	Alibi off	Catch outside
3+2, Framer	Soft	0.5294 ± 0.0200	0.5306 ± 0.0198	0.5294 ± 0.0200
3+2, Framer	Sharp	0.5124 ± 0.0202	0.5172 ± 0.0209	0.5110 ± 0.0204
5+2, Truthful	Mean	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
5+2, Truthful	Soft	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
5+2, Truthful	Sharp	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
5+2, Lone liar	Mean	0.1326 ± 0.0181	0.1876 ± 0.0171	0.1326 ± 0.0181
5+2, Lone liar	Soft	0.1598 ± 0.0184	0.2064 ± 0.0199	0.1316 ± 0.0189
5+2, Lone liar	Sharp	0.2792 ± 0.0187	0.3244 ± 0.0154	0.1316 ± 0.0189
5+2, Alibi	Mean	0.3122 ± 0.0198	0.2312 ± 0.0224	0.3122 ± 0.0198
5+2, Alibi	Soft	0.3802 ± 0.0242	0.2310 ± 0.0215	0.2890 ± 0.0183
5+2, Alibi	Sharp	0.4198 ± 0.0282	0.3414 ± 0.0191	0.2676 ± 0.0210
5+2, Framer	Mean	0.4060 ± 0.0248	0.5244 ± 0.0236	0.4060 ± 0.0248
5+2, Framer	Soft	0.4014 ± 0.0275	0.4890 ± 0.0259	0.4010 ± 0.0276
5+2, Framer	Sharp	0.4054 ± 0.0214	0.4706 ± 0.0229	0.3790 ± 0.0261
7+2, Truthful	Mean	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
7+2, Truthful	Soft	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
7+2, Truthful	Sharp	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
7+2, Lone liar	Mean	0.0908 ± 0.0179	0.1416 ± 0.0139	0.0908 ± 0.0179
7+2, Lone liar	Soft	0.1438 ± 0.0231	0.1834 ± 0.0169	0.0902 ± 0.0179
7+2, Lone liar	Sharp	0.3012 ± 0.0237	0.3672 ± 0.0205	0.0902 ± 0.0179
7+2, Alibi	Mean	0.2080 ± 0.0190	0.1632 ± 0.0191	0.2080 ± 0.0190
7+2, Alibi	Soft	0.3666 ± 0.0250	0.2090 ± 0.0157	0.1916 ± 0.0193
7+2, Alibi	Sharp	0.4362 ± 0.0256	0.4012 ± 0.0254	0.1786 ± 0.0199
7+2, Framer	Mean	0.2654 ± 0.0176	0.4670 ± 0.0212	0.2654 ± 0.0176
7+2, Framer	Soft	0.2574 ± 0.0180	0.3906 ± 0.0212	0.2568 ± 0.0176
7+2, Framer	Sharp	0.3162 ± 0.0163	0.4192 ± 0.0231	0.2412 ± 0.0172

Mean catch-outside false-ejection outcomes differ from baseline only in two 3+2 alibi episodes (0.5176 versus 0.5180): replays locate a 2.22×10^{-16} change in one candidate score that resolves a floating-point argmax tie and changes the meeting outcome. Algebraic equality therefore does not imply bitwise equality of decisions.

In the alibi condition, soft credibility exceeds the baseline mean’s false-ejection rate by paired differences of 0.0112 ± 0.0080 , 0.0680 ± 0.0136 , and 0.1586 ± 0.0112 at 3+2, 5+2, and 7+2. Moving the caught penalty outside reduces soft alibi false ejections by 0.0242 ± 0.0066 , 0.0912 ± 0.0150 , and 0.1750 ± 0.0084 relative to its own baseline. Removing alibi credit also lowers alibi false ejections for every tested rule and size. These interventions support a role for alibi credit and catch-penalty placement in the observed behavior, while allowing responses and resulting testimony use to change.

The improvements are conditional on the attack and size. Removing alibi credit raises lone-liar and framer false ejections; for soft credibility under the framer the increases are 0.0876

± 0.0143 at 5+2 and 0.1332 ± 0.0222 at 7+2. Moving the caught penalty outside raises the soft lone-liar rate at 3+2 by 0.1540 ± 0.0188 , although it reduces lone-liar false ejections at the larger sizes. Soft framer rates change by at most 0.0006 under catch-outside. Sharper weighting also has no uniform advantage: at 7+2 its baseline false-ejection rate exceeds the mean by 0.2104 ± 0.0139 for the lone liar, 0.2282 ± 0.0172 for alibi, and 0.0508 ± 0.0124 for the framer.

False-ejection reductions can reflect abstention or tied votes rather than more correct ejections. Under coordinated alibi at 3+2, every tested rule/variant has creator-ejection rate zero and coalition-favorable rate one; reducing false ejections only increases no-ejection outcomes. Under the lone liar at 5+2, soft catch-outside lowers false ejection from 0.1598 to 0.1316, but raises no-ejection from 0.0402 ± 0.0049 to 0.1100 ± 0.0105 and coalition-favorable outcomes from 0.2000 ± 0.0207 to 0.2416 ± 0.0257 . Thus these measurements distinguish testimony-based alibi support and catch-penalty placement from increased coalition weight, without establishing a unique causal explanation or an improvement under every objective.

6.2.2 A principled baseline: hypothesis elimination

The hypothesis-scoring crew of Section 3.4 reduces alibi false ejections to 0.156 at 5+2 and 0.084 at 7+2: reductions of 53% and 59% relative to the mean. The improvement over the mean holds in every seed. Creator ejection is comparable to the best other rule at 5+2 (0.36 versus 0.37 for dependence-aware credibility) and highest at 7+2 (0.62). It also has the lowest observed false-ejection rate against the framer at 5+2 and 7+2 and against the lone liar at 7+2.

The procedure scores jointly consistent subsets rather than adding speaker suspicion rows. However, this archived version includes coalition claims in some mixed-honesty branches and gives nonzero smoothed weight to otherwise excluded hypotheses. Its scripted performance therefore provides empirical evidence for this implementation, not a guarantee of immunity to coordinated alibis.

With truthful testimony, the hypothesis crew ejects an innocent in 0.077 of meetings at 5+2 and 0.125 at 7+2, while the mean and credibility rules have no observed false ejections. The heuristics can exploit truthful self-placement by the creator, whereas the hypothesis rule partially discounts adversarial claims and uses a simplified incident-location model. Its residual errors can reflect both ambiguous evidence and model misspecification; they do not directly measure an irreducible information limit. At 3+2 it performs worst against the lone liar. Section 6.9 tests whether its empirical advantage against the scripted alibi survives a trained adversary.

6.3 F3: single-meeting crossplay depends on opponent and population

All 30 corrected F3 jobs are complete, with 400 PPO updates in each of the C0, D1 and C1 stages. The learned spatial policies now receive their full within-round private movement and sighting histories. Crossplay uses fixed episode indices, isolated policy-sampling streams

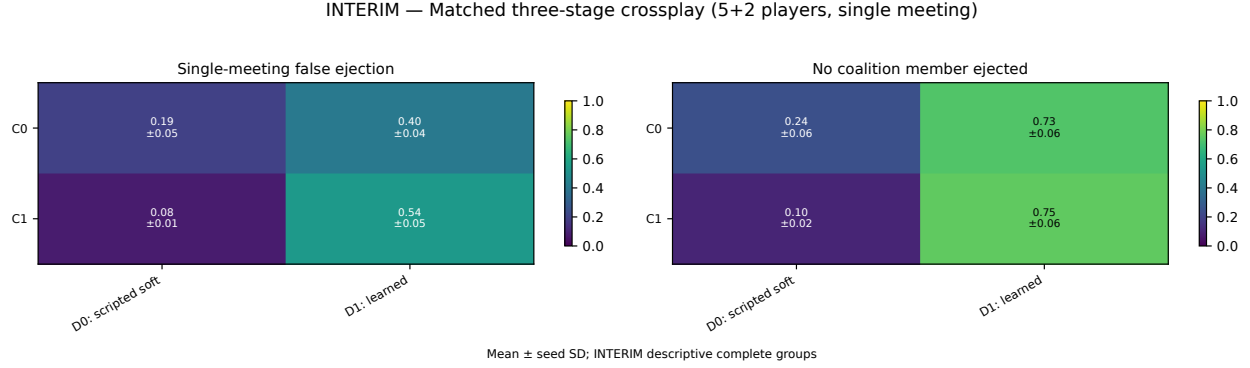


Figure 3. Corrected single-meeting crossplay at 5+2, showing the full two-by-two pairing of coalition policies C0/C1 and defenses D0/D1. D0 is scripted soft credibility; D1 is trained against frozen C0, and C1 is trained against frozen D1. Entries are mean $\pm$ sample SD across ten training seeds, with 1,000 final holdout episodes per pair and seed. The right panel’s “no coalition member ejected” outcome includes skips. Table 7 gives all four pairings at all three populations. These are descriptive complete-group results, with inference deferred until the full planned study is complete.

and the independent final evidence stream. Every game has one meeting, including when the coalition creates no incident. Table 7 therefore reports both unconditional harm and the observed conditional rates reconstructed from all 120,000 retained game records.

The corrected results change the archived incident-avoidance account. Against D0, C0 creates an incident in 0.986, 0.963 and 0.943 of episodes at 3+2, 5+2 and 7+2. When the same frozen C0 instead faces D1, the incident rates fall to 0.087, 0.254 and 0.321. This change follows from the changed joint trajectories and does not represent additional coalition learning. The conditional columns describe different resulting evidence populations, so their differences do not isolate a testimony mechanism.

Against frozen C0, D1 has a higher observed mean false-ejection rate than D0 at every population: 0.188 to 0.463, 0.194 to 0.403, and 0.211 to 0.475. Coalition retraining then has a population-dependent result. Against the same D1 checkpoint, C1 raises mean FE from 0.403 to 0.538 at 5+2 and from 0.475 to 0.587 at 7+2, but lowers it from 0.463 to 0.415 at 3+2. The latter result does not support a universal counter-adaptation gain in false ejection.

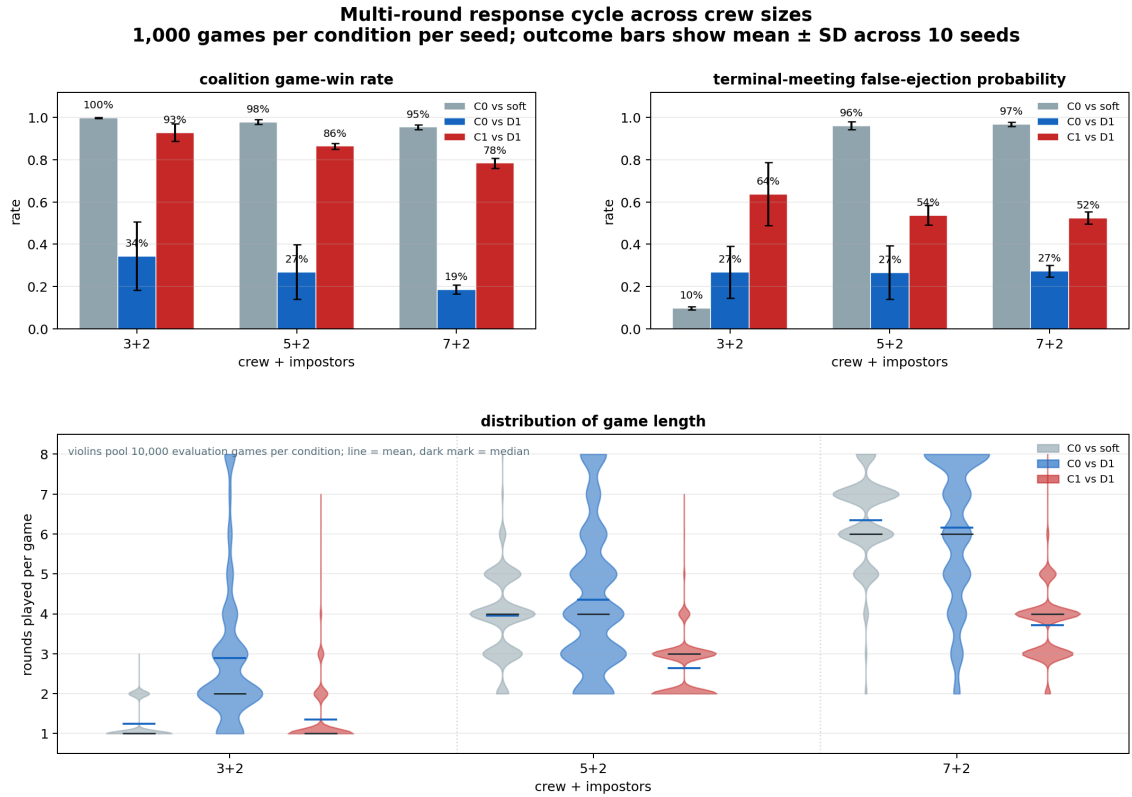
The complete crossplay also limits the transfer claim. Returning C1 to D0 gives FE rates of 0.068, 0.076 and 0.083, below C0’s 0.188, 0.194 and 0.211 against D0. Thus the policy trained against D1 produces less innocent-player harm against the scripted reference in these runs. Together these comparisons describe opponent-dependent outcomes at the tested fixed training budget. They establish neither an equilibrium nor a general ordering of learned and scripted defenses. The full-study paired uncertainty and multiplicity analysis remains pending.

Table 7. Corrected single-meeting outcomes, mean $\pm$ sample SD of within-seed rates. D0 denotes the scripted soft-credibility crew. Incident and FE are per-episode probabilities; the three conditional columns use the indicated stratum. All unconditional entries use ten seeds and 1,000 episodes per seed and pair. $\dagger$ At 3+2, C0/D1 has incidents in 6/10 seeds; $\ddagger$ C1/D1 has incidents in 8/10 seeds. Their incident-conditional means and SDs use those six or eight defined seed rates. All other conditional entries use ten seeds. Empty strata remain undefined and are not imputed as zero risk; all seeds remain in the unconditional summaries. Some incident-free strata are small (as few as five episodes in a seed). Conditional means are descriptive averages of the observed strata, not causal effects of creating an incident, and need not satisfy a pooled mixture identity with the unconditional seed means.

Population, pair	Incident	FE	FE given incident	FE given no incident	Creator ejected given incident
3+2, C0/D0	0.986 $\pm$ 0.005	0.188 $\pm$ 0.045	0.183 $\pm$ 0.045	0.589 $\pm$ 0.227	0.130 $\pm$ 0.050
3+2, C0/D1 $\dagger$	0.087 $\pm$ 0.140	0.463 $\pm$ 0.073	0.276 $\pm$ 0.236	0.486 $\pm$ 0.059	0.140 $\pm$ 0.161
3+2, C1/D1 $\ddagger$	0.302 $\pm$ 0.286	0.415 $\pm$ 0.089	0.405 $\pm$ 0.207	0.414 $\pm$ 0.090	0.062 $\pm$ 0.050
3+2, C1/D0	0.949 $\pm$ 0.089	0.068 $\pm$ 0.034	0.050 $\pm$ 0.018	0.458 $\pm$ 0.091	0.364 $\pm$ 0.113
5+2, C0/D0	0.963 $\pm$ 0.027	0.194 $\pm$ 0.051	0.175 $\pm$ 0.041	0.661 $\pm$ 0.091	0.587 $\pm$ 0.042
5+2, C0/D1	0.254 $\pm$ 0.178	0.403 $\pm$ 0.040	0.353 $\pm$ 0.082	0.416 $\pm$ 0.040	0.201 $\pm$ 0.105
5+2, C1/D1	0.788 $\pm$ 0.132	0.538 $\pm$ 0.053	0.553 $\pm$ 0.055	0.480 $\pm$ 0.047	0.119 $\pm$ 0.030
5+2, C1/D0	0.984 $\pm$ 0.003	0.076 $\pm$ 0.014	0.067 $\pm$ 0.014	0.651 $\pm$ 0.138	0.696 $\pm$ 0.036
7+2, C0/D0	0.943 $\pm$ 0.030	0.211 $\pm$ 0.022	0.188 $\pm$ 0.019	0.604 $\pm$ 0.103	0.619 $\pm$ 0.028
7+2, C0/D1	0.321 $\pm$ 0.216	0.475 $\pm$ 0.037	0.474 $\pm$ 0.035	0.476 $\pm$ 0.047	0.111 $\pm$ 0.020
7+2, C1/D1	0.754 $\pm$ 0.117	0.587 $\pm$ 0.040	0.614 $\pm$ 0.051	0.516 $\pm$ 0.039	0.080 $\pm$ 0.017
7+2, C1/D0	0.970 $\pm$ 0.010	0.083 $\pm$ 0.020	0.066 $\pm$ 0.019	0.611 $\pm$ 0.108	0.732 $\pm$ 0.030

6.4 F4: multi-round play creates a moving response cycle

The multi-round extension changes the interpretation of a meeting. Agents move, create an incident, complete tasks, report, make claims, respond, vote, and continue if neither side has met the terminal condition. A learned defender can affect the ejection and the amount of future evidence available in the game, and here the learned defender is decisive.



The 3+2 setting is a parity-boundary case: one successful kill leaves two impostors versus two crew.

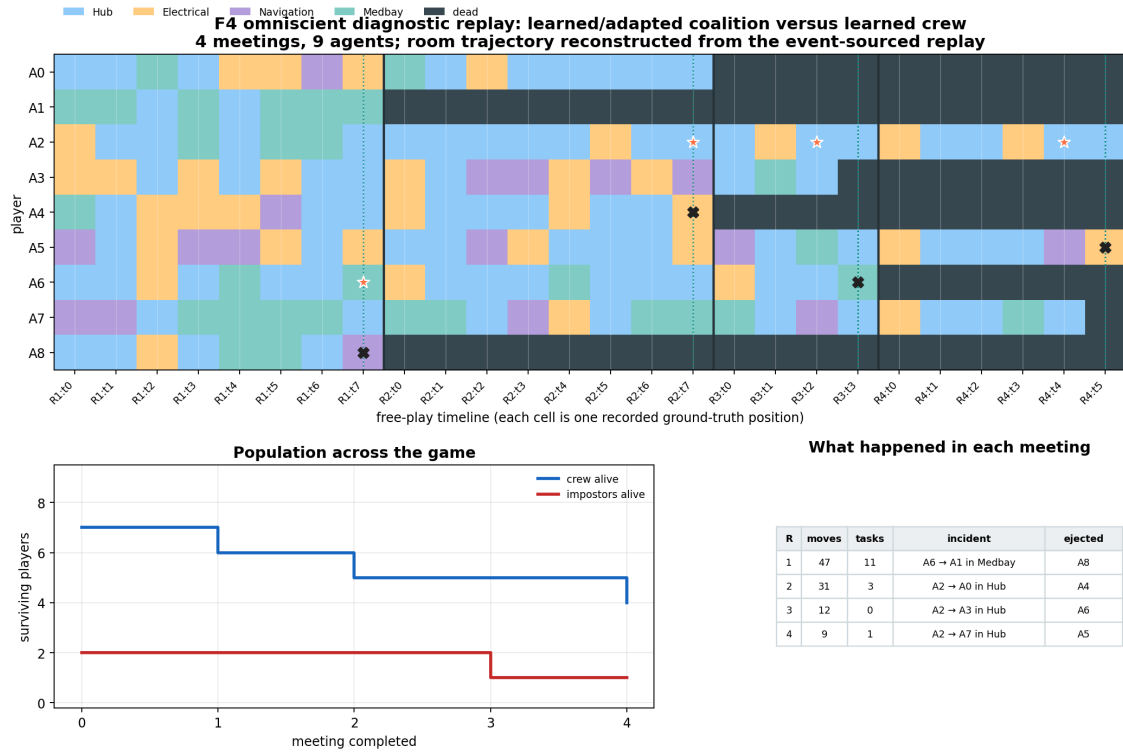
Figure 4. Archived measurements. Multi-round outcomes and game-length distributions. Each condition uses 1,000 evaluation games per seed and ten seeds; the pooled game-length distributions are descriptive.

Table 8. Archived measurements. Multi-round response cycle: coalition game-win rate, terminal-meeting false-ejection probability (FE, one final meeting per game), and mean rounds played, means across ten seeds.

Crew + impostors	C0 vs scripted crew	C0 vs learned D1	C1 vs learned D1
3 + 2	99.9% coalition wins; FE 0.10; 1.24 rounds	34.4%; 0.27; 2.89	92.9%; 0.64; 1.35
5 + 2	97.9%; FE 0.96; 3.95 rounds	26.9%; 0.27; 4.35	86.4%; 0.54; 2.64
7 + 2	95.5%; FE 0.97; 6.34 rounds	18.6%; 0.27; 6.16	78.4%; 0.53; 3.71

Against the scripted crew the learned coalition wins almost every multi-round game and, at 5+2 and 7+2, ends with an innocent ejection in 96 to 97 percent of games. This archived FE metric inspects only the final meeting; it is neither a per-meeting average nor cumulative harm across rounds. The same scripted crew that is a strong single-meeting defender is helpless once the coalition controls movement and the incident: movement and incident creation change both the available evidence and the opportunities to win. The current measurements do not isolate which of those mechanisms causes the difference. The learned defender D1 cuts the coalition’s win rate by 0.65 to 0.77 at every crew size ($p = 0.002$). Terminal-meeting FE falls to about 0.27 at 5+2 and 7+2, but rises from 0.10 to 0.27 at 3+2. It lengthens the game at 3+2 and 5+2. The adapted coalition C1 recovers most of the loss (win rates 0.93, 0.86 and 0.78) and shortens the game again to 1.4, 2.6 and 3.7 rounds. The 3+2 setting is a parity-boundary case: one successful kill leaves two impostors against two crew, so it is inherently easier for the coalition.

A separate reaggregation of the saved episode records gives the probability of at least one incident-bearing innocent ejection anywhere in the game: for C0 versus scripted, C0 versus D1, and C1 versus D1 respectively, it is 0.074, 0.066 and 0.313 at 3+2; 0.108, 0.149 and 0.832 at 5+2; and 0.088, 0.166 and 0.931 at 7+2. This cumulative quantity excludes incident-free ejections because the original accumulator was incident-gated. Its different pattern shows why a decrease in terminal-meeting FE cannot be interpreted as a decrease in whole-game harm.



Omniscient diagnostic; direct role labels suppressed. Outcome: coalition-favourable; * incident, X ejection, teal line report

Figure 5. Archived measurements. Omniscient diagnostic view of one illustrative archived full-game replay, with direct role labels suppressed. It displays ground-truth trajectories, incident creators and victims, and role-specific survivor counts; these are not all available to any player’s private view.

6.5 Controls: what the coalition’s coordination depends on, and whether budget or reward explains the last-mover advantage

Coordination ablation. The C0-stage ablations retain strong coalition performance (Figure 6, Table 9): multi-round wins average 0.96 to 0.97 across the four conditions, and same-target voting averages 0.18 to 0.19 in the single meeting and 0.36 to 0.52 in multi-round play. None of the tested paired differences is significant at 0.05 (minimum exact sign-flip $p = 0.063$), but five seeds provide limited power and no equivalence margin was specified. The result concerns removal of an explicit symbol channel and partner-identity observation block. Role-dependent incident-action masks still supply information about possible targets, and shared policies and objectives remain. This is therefore not a test of complete partner ignorance, independent incentives, or whether a full adaptation lineage would persist without these features.

Table 9. Archived measurements. Coordination ablation, C0 stage at 5+2 against the scripted crew; mean $\pm$ SD across five seeds, 1,000 evaluation episodes per seed.

Regime, metric	Channel + partner id	No channel	No partner id	Neither
Single meeting, false ejection	0.39 $\pm$ 0.08	0.40 $\pm$ 0.09	0.34 $\pm$ 0.04	0.40 $\pm$ 0.04
Single meeting, same-target voting	0.19 $\pm$ 0.05	0.19 $\pm$ 0.05	0.18 $\pm$ 0.03	0.18 $\pm$ 0.03
Single meeting, real creator survives (per episode)	0.09 $\pm$ 0.08	0.04 $\pm$ 0.03	0.09 $\pm$ 0.05	0.04 $\pm$ 0.02
Multi-round, coalition wins	0.97 $\pm$ 0.02	0.96 $\pm$ 0.01	0.97 $\pm$ 0.01	0.97 $\pm$ 0.01
Multi-round, same-target voting	0.43 $\pm$ 0.07	0.44 $\pm$ 0.08	0.36 $\pm$ 0.17	0.52 $\pm$ 0.09

Defender budget. The tested increases in defender updates do not remove the archived last-mover gain (Figure 7, Table 10). In the single meeting a defender trained for 800 or 1,600 updates instead of 400 is no better against C0 (false ejection 0.49 and 0.46 against 0.44), and C1, trained for 400 updates against each of them, reaches the same 0.58. In the multi-round game the extra budget does help the defender stage: D1 with 800 updates holds the coalition to a 0.13 win rate instead of 0.29. The coalition trained against that stronger defender still wins about 0.86 of games against both defenders; this numerical similarity is not an equivalence result. These controls vary update count, not information access or the rollout implementation.

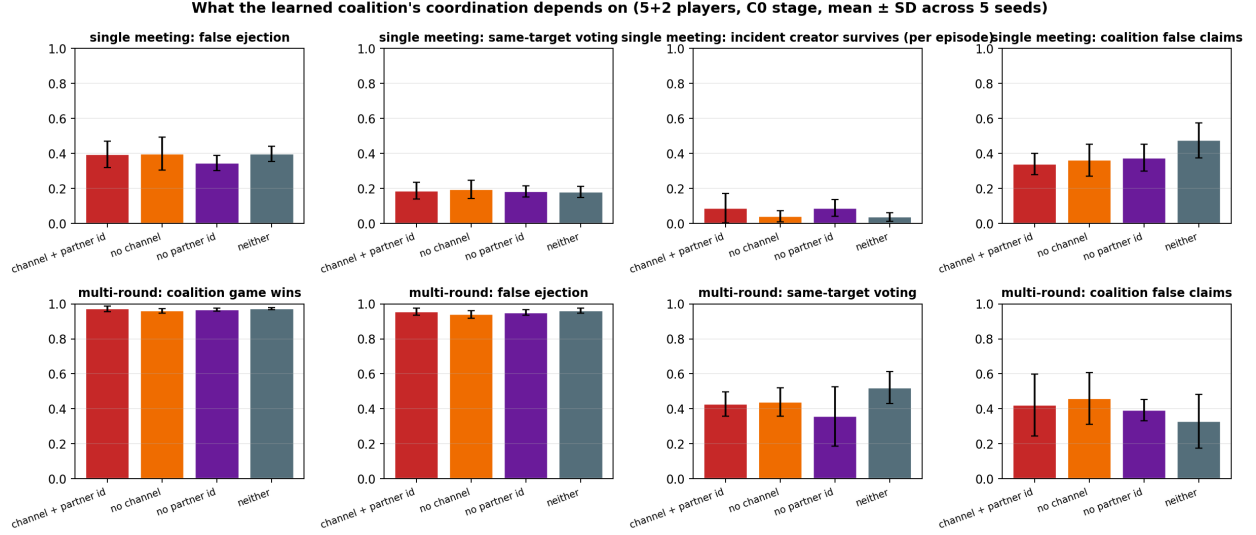


Figure 6. Archived measurements. Coordination ablation: the C0 stage trained and evaluated under four information conditions (5+2 players, mean $\pm$ SD across five seeds).

Table 10. Archived measurements. Defender budget at 5+2: false-ejection rate (single meeting) or coalition game-win rate (multi-round), mean $\pm$ SD across the same five seeds; C0 and C1 always train for 400 updates.

Regime, defender updates	C0 vs scripted	C0 vs D1	C1 vs D1
Single meeting, 400 (reference)	0.39 $\pm$ 0.09	0.44 $\pm$ 0.02	0.59 $\pm$ 0.04
Single meeting, 800	0.39 $\pm$ 0.10	0.49 $\pm$ 0.05	0.58 $\pm$ 0.04
Single meeting, 1,600	0.40 $\pm$ 0.10	0.46 $\pm$ 0.03	0.58 $\pm$ 0.03
Multi-round, 400 (reference)	0.98 $\pm$ 0.01	0.29 $\pm$ 0.14	0.86 $\pm$ 0.02
Multi-round, 800	0.98 $\pm$ 0.01	0.13 $\pm$ 0.15	0.86 $\pm$ 0.03

Reward specification. Positive last-mover gains persist under the tested objectives, while some outcome levels change (Figures 8 and 9, Table 11). All rows here use the same five seeds. In the single meeting, the coalition-stage rates are 0.756, 0.766 and 0.759 under the reference, shared balanced-ejection and shared meeting-ejection objectives. The respective last-mover gains are $+0.19 \pm 0.03$, $+0.14 \pm 0.05$ and $+0.20 \pm 0.04$. Defender-stage rates are 0.562, 0.627 and 0.564; the balanced objective increases this rate by $+0.065 \pm 0.038$ relative to the reference ($p = 0.063$). Similar point estimates and non-significant contrasts do not establish that the reward specifications are interchangeable.

Generation 0 also differs: coalition-favorable rates are 0.773, 0.675 and 0.965. With the meeting-ejection objective, incident-free meetings carry zero reward regardless of the tally; this does not prevent a coalition member from being ejected. The different machines used

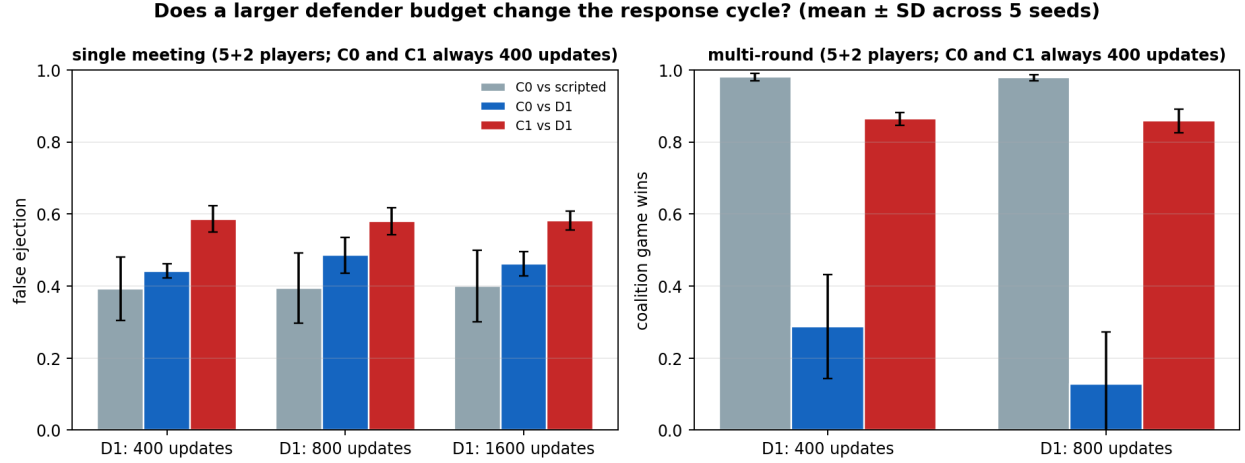


Figure 7. Archived measurements. Defender budget: the response cycle with the defender trained for 400, 800 and 1,600 updates (5+2 players, mean $\pm$ SD across five seeds).

for some reference and control runs are another possible source of variation.

In multi-round play, the framing-count reward counts innocent ejections only in incident-bearing rounds and pays neither side directly for winning. Coalition-stage win rates are 0.836 under framing count and 0.849 under survival, but defender-stage rates are 0.660 and 0.336. Last-mover gains are $+0.17 \pm 0.09$ and $+0.52 \pm 0.15$, positive in every seed. These are distinct levels, not invariance to reward specification. The plotted terminal-meeting FE metric is also different from the cumulative, incident-gated framing count actually optimized. As with the other spatial learning results, these controls require reruns after the collector repair.

Table 11. Archived measurements. Reward control at 5+2: the primary metric (coalition-favorable meeting rate in the single meeting, coalition game-win rate in the multi-round game) after the coalition stages and after the defender stages of the loop, and the last-mover gain from cross-play; mean $\pm$ SD across the same five seeds, the reference rows truncated to the control’s horizon.

Regime, objective	Generation 0	After coalition stages	After defender stages	Last-mover gain
Single meeting, side-specific objectives (paper), $k = 4$	0.77	0.756 ± 0.009	0.562 ± 0.031	$+0.19 \pm 0.03$
Single meeting, both sides balanced ejection	0.67	0.766 ± 0.012	0.627 ± 0.039	$+0.14 \pm 0.05$
Single meeting, both sides meeting ejection	0.96	0.759 ± 0.024	0.564 ± 0.037	$+0.20 \pm 0.04$
Multi-round, survival (paper), $k = 2$	0.97	0.849 ± 0.023	0.336 ± 0.140	$+0.52 \pm 0.15$
Multi-round, both sides framing count	0.33	0.836 ± 0.020	0.660 ± 0.089	$+0.17 \pm 0.09$

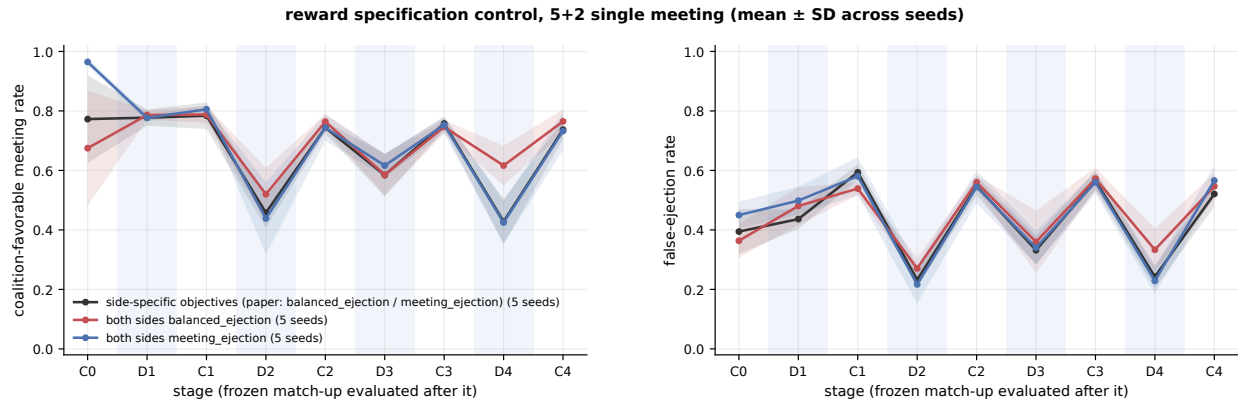


Figure 8. Archived measurements. Reward control, single meeting: the loop under the paper’s side-specific objectives and under one objective for both sides (mean $\pm$ SD across five seeds).

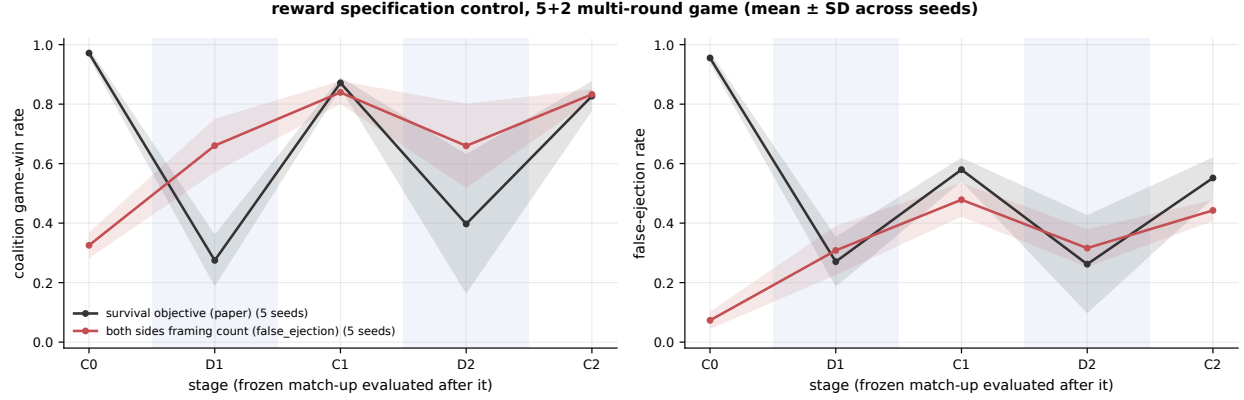


Figure 9. Archived measurements. Reward control, multi-round game: the first five stages of the loop under the survival objective and under the framing-count objective on both sides (mean $\pm$ SD across five seeds).

6.6 F5: repeated outcome reversals over the observed training horizon

The first three stages of the multi-round loop use the F4 protocol and standard budget and reproduce it within training noise (coalition game-win rate 0.97 against the scripted crew, 0.25 against D1, 0.88 for C1 against D1; F4: 0.98, 0.27 and 0.86; the loop’s seeds 5 to 9 were trained on a different machine). The remaining generations show a persistent alternation (Figure 10, Table 12). The coalition-stage level over the last three generations is 0.82 and the defender-stage level 0.37; the least-squares slopes over generations are -0.003 and $+0.008$ per generation; the sawtooth amplitude is 0.50 over the first three generations and 0.45 over the last three; and the late-minus-early change of the coalition-stage win rate is -0.016 ± 0.021 across seeds (exploratory sign-flip $p = 0.049$). This small decrease does not establish stationarity; the alternation remains large over the observed horizon. Terminal-meeting false ejection shows a similar alternation (0.52 after coalition stages, 0.23 after defender stages), as do game length (2.7 rounds after coalition stages, 3.0 after defender stages) and both sides’ policy entropies (no trend). The other configurations show the same picture at their own levels.

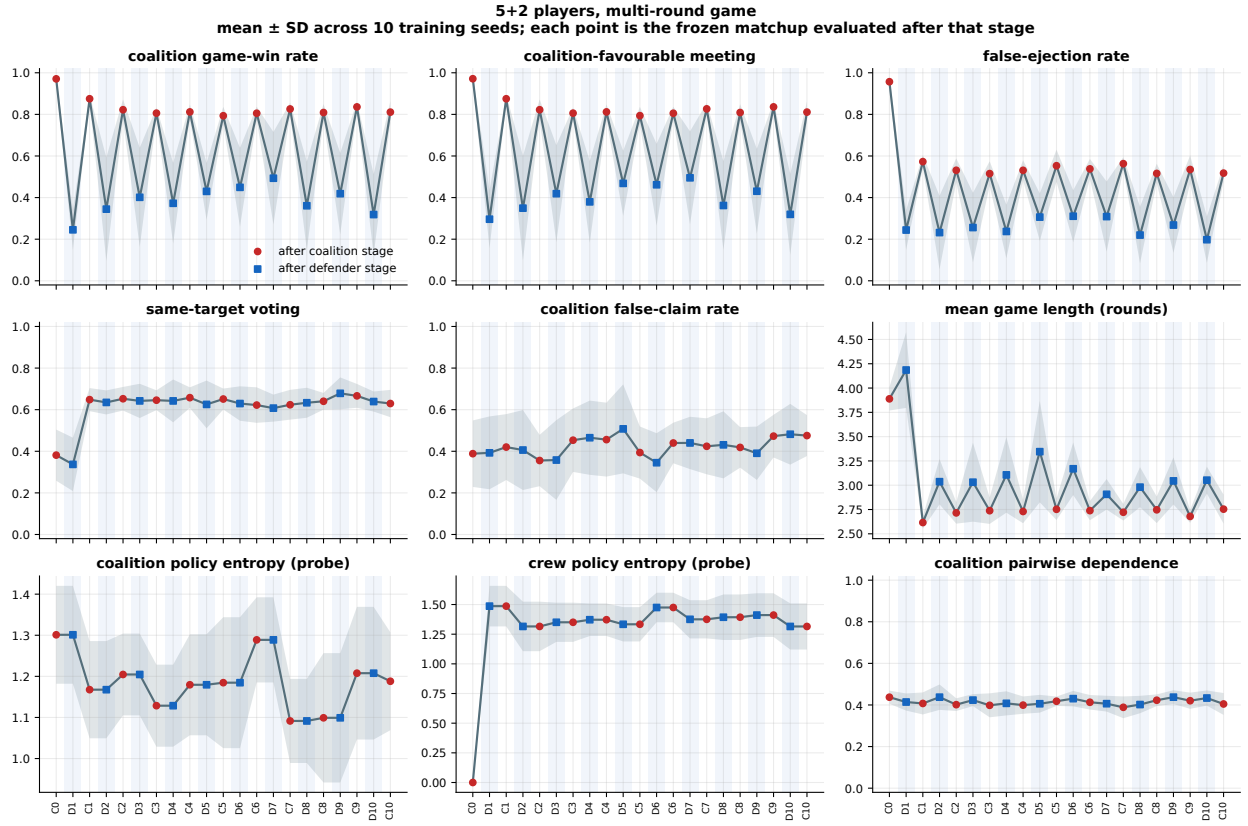


Figure 10. Archived measurements. The ten-generation loop for 5+2 players in the multi-round game. Each point is the frozen match-up evaluated after that stage (red: after a coalition stage, blue: after a defender stage); bands are $\pm$ SD across ten training seeds.

Table 12. Archived measurements. Late-horizon levels of the primary metric (mean over the last three generations; the coalition game-win rate in the multi-round game and the coalition-favorable meeting rate in the single meeting), the sawtooth amplitude over the first and last three generations (first and last generation for $k = 3$), and the last-mover gain (C_i against D_i minus C_{i-1} against D_i in cross-play, mean $\pm$ SD across seeds; positive in every seed of every configuration, exact sign-flip $p = 0.002$ at 5+2 multi-round).

Configuration	After coalition stages	After defender stages	Sawtooth early $\rightarrow$ late	Last-mover gain
5+2, multi-round, $k = 10$ (10 seeds)	0.82	0.37	0.50 $\rightarrow$ 0.45	+0.43 $\pm$ 0.07
5+2, single meeting, $k = 10$ (10 seeds)	0.76	0.49	0.17 $\rightarrow$ 0.27	+0.24 $\pm$ 0.03
3+2, multi-round, $k = 10$ (10 seeds)	0.95	0.35	0.56 $\rightarrow$ 0.59	+0.58 $\pm$ 0.08
7+2, multi-round, $k = 3$ (5 seeds)	0.75	0.40	0.59 $\rightarrow$ 0.26	+0.35 $\pm$ 0.05

Cross-play. The cross-play matrix (Figure 11) has one strong structure and one weak one. The strong one is the last-mover advantage: every C_i beats its own D_i by 0.43 on average more than C_{i-1} did, in every seed, and the diagonal averages 0.82 from C_1 onward. Defenders do not accumulate strength: a defender’s average win rate conceded to all coalitions is 0.54 to 0.69 with no trend in the generation, and a coalition’s average against all learned defenders is 0.60 to 0.63 from C_1 onward. A coalition’s average win rate against the defenders that came after it equals its average against those that came before it (0.54 and 0.54); the ordering of generations carries no information beyond adjacency. The weak structure is parity: the earlier version of this study reported that odd coalitions beat odd defenders and lose to even ones. Measured properly, as a coalition’s win rate against non-adjacent defenders of the same parity minus the opposite parity, the gap is $+0.09 \pm 0.14$ in the 5+2 multi-round game (positive in 80 percent of seeds, $p = 0.076$), $+0.14 \pm 0.14$ at 3+2 (80 percent, $p = 0.023$), and $+0.07 \pm 0.03$ in the 5+2 single meeting (100 percent, $p = 0.002$). These finite-sample parity contrasts do not establish a period-2 cycle in strategy space.

Probe-set distances. Independently trained successive policies differ on the scripted probe set. The Kullback–Leibler divergence from a policy to its successor is 5.6 nats for the coalition and 5.0 for the crew (summed over heads) in the last three generations and just as large in the first three; the Jensen–Shannon divergence is 0.96 and 0.91 bits; greedy actions differ in 86 to 89 percent of probe states; and the crew moves its voting pattern about twice as far

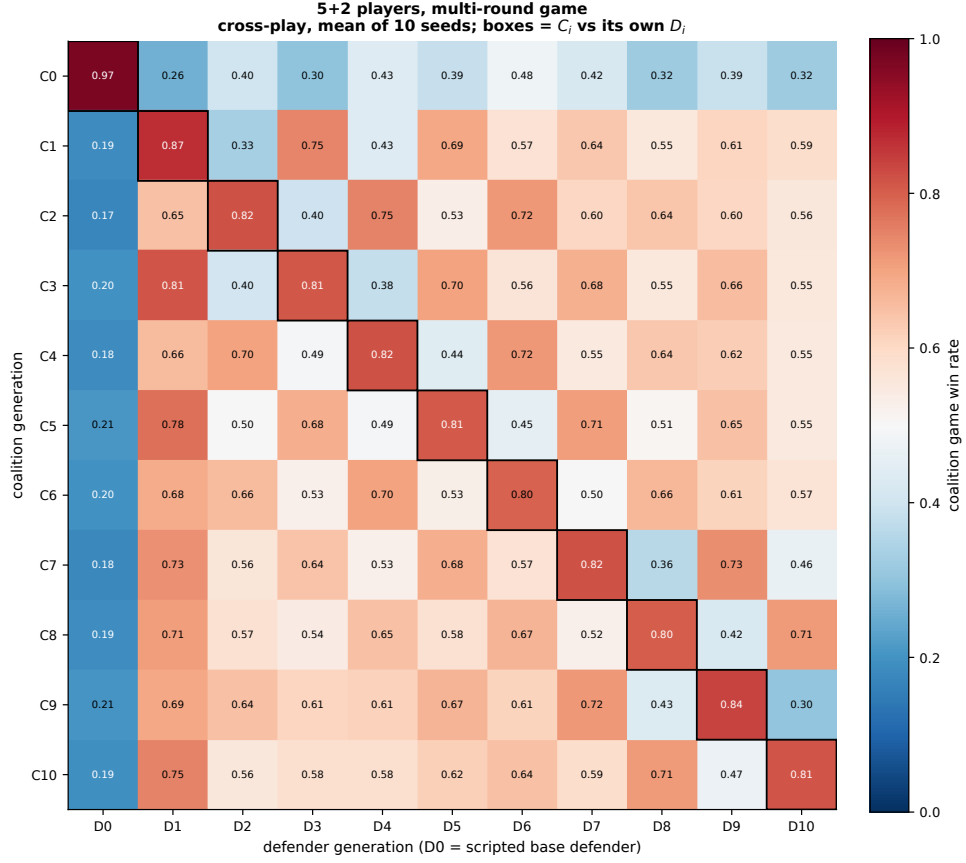


Figure 11. Archived measurements. Cross-generation play: coalition game-win rate of every C_i (rows) against every D_j (columns), 5+2 players, multi-round game, mean of ten seeds. Boxes mark C_i against its own D_i .

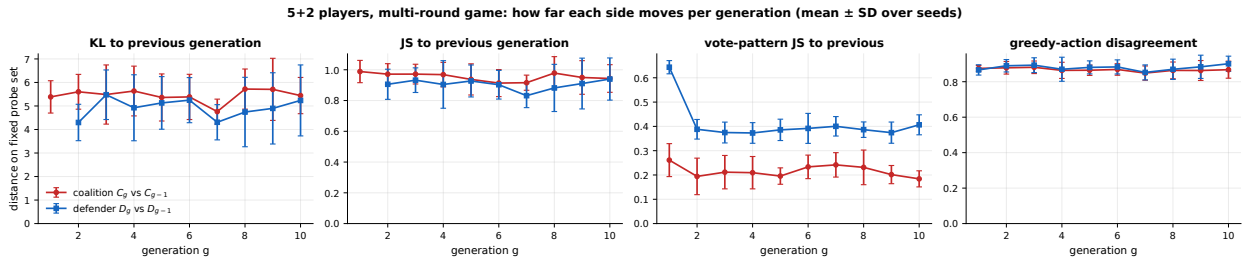


Figure 12. Archived measurements. Distance between successive same-side policies on the fixed probe set, 5+2 players, multi-round game (mean $\pm$ SD across ten seeds).

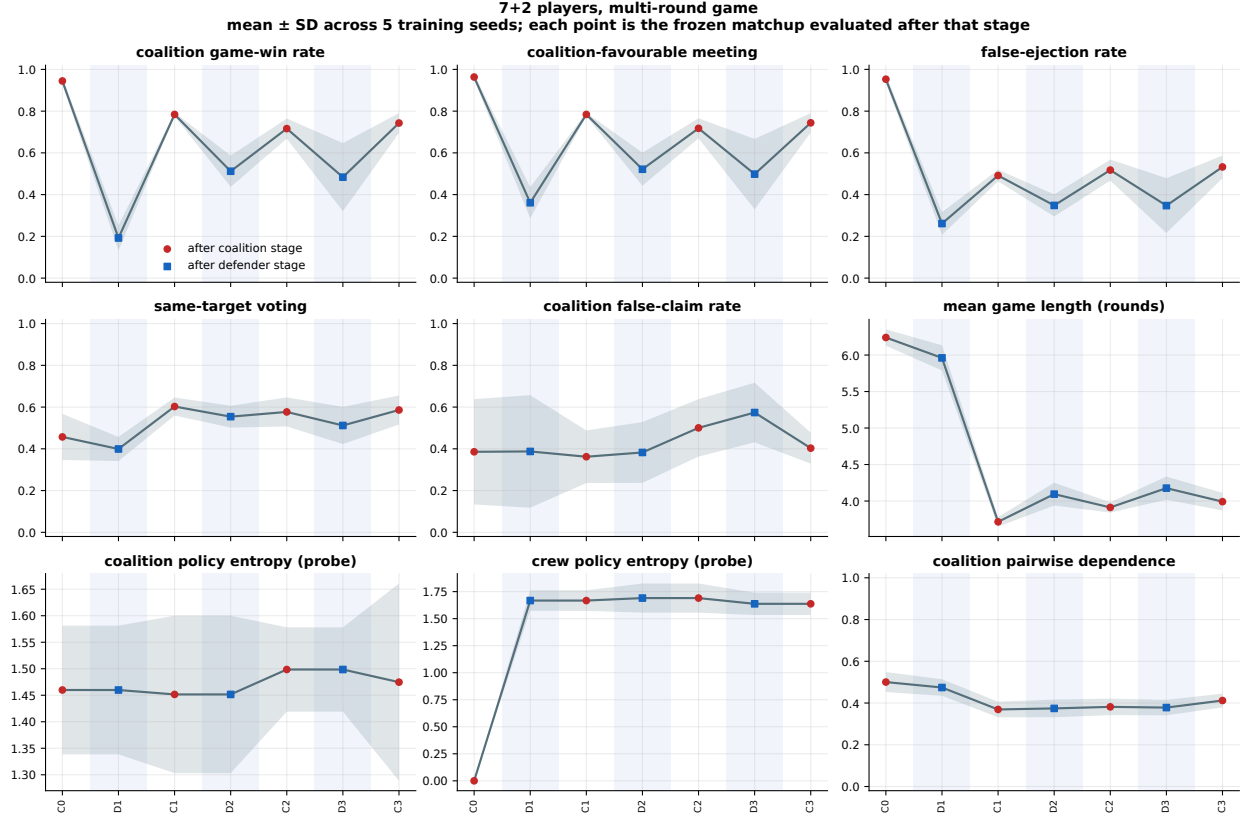


Figure 13. Archived measurements. The loop for 7+2 players in the multi-round game, three generations, mean $\pm$ SD across five training seeds.

per generation as the coalition (vote-pattern JS 0.39 against 0.21). The distances remain substantial at late generations. Because each stage starts from a fresh initialization and probes are drawn from scripted rather than learned play, this does not test convergence of a continuously updated policy or exclude restart variability as a contributor.

What the lineage learns. The coalition’s false-claim rate is 0.39 against the scripted crew and 0.43 to 0.46 in the loop; learned defenders do not catch lies the way the scripted rules do, so lying stays cheap, and every later coalition is weaker against the scripted crew than C_0 was (cross-play column D_0 : 0.19 on average against 0.97 for C_0). Vote coordination is acquired in the first generation and kept (same-target voting 0.38 $\rightarrow$ 0.65). The archived dependence statistic uses post-ejection survivors and is not a valid estimate of the full pre-vote coalition population; its apparent stability is not used as mechanism evidence.

The largest crew size. At 7+2 (Figure 13, five seeds) the alternation is present with the same sign: the coalition wins 0.94, 0.78, 0.72 and 0.74 after its stages and 0.19, 0.51 and 0.48 after the defender stages, with a last-mover gain of $+0.35 \pm 0.05$ (positive in every seed). Over the three generations run, the defender stages weaken and the amplitude shrinks from 0.59 to 0.26, so the 7+2 loop looks more like a damped alternation than the persistent one of 5+2. Three generations do not settle whether it damps out; what the run shows is that the last-mover advantage and the coalition’s stage-level lead are not specific to the smaller

crews.

6.7 F6: an analytical model of corroboration capture under correlated testimony

The model isolates the one failure mode that the framing of “credibility collapse” usually refers to: manufactured agreement earning a coalition more than its share of a trust rule’s weight. We use the term in that sense only; Section 6.2’s archived scripted diagnostics do not show average coalition-weight amplification in those conditions. The model replaces claims and votes by real-valued reports about a latent state θ . n honest agents report $x_i = \theta + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ i.i.d.; m coalition agents report $y_j = \theta + b + \eta_j$ with equicorrelated Gaussian noise of marginal variance σ_c^2 , independent of honest noise, and pairwise correlation ρ ($\rho = 1$: identical, fully coordinated reports); $f = m/(n + m)$ is the coalition fraction. The corroboration count $c_i = \#\{k \neq i : |r_i - r_k| \leq h\}$ is the analogue of counting speakers whose claims are jointly satisfiable with one’s own, and the credibility-weighted estimate is $\hat{\theta} = \sum_i w_i r_i$ with $w_i \propto e^{\gamma c_i}$, gain $\gamma = 1/\tau$. Because differences of Gaussians are Gaussian, the pairwise agreement probabilities are explicit, $p_{HH} = \text{erf}(h/2\sigma)$, $p_{CC}(\rho) = \text{erf}(h/(2\sigma_c\sqrt{1-\rho}))$, $p_{CC}(1) = 1$, $p_{HC}(b) = \Phi(\frac{h-b}{s}) - \Phi(\frac{-h-b}{s})$ with $s^2 = \sigma^2 + \sigma_c^2$, and the expected corroboration counts are $\bar{c}_H = (n-1)p_{HH} + m p_{HC}$ and $\bar{c}_C = (m-1)p_{CC}(\rho) + n p_{HC}$. In mean field the coalition’s total weight is the logistic

$$W_C(\gamma, \rho, f) = \frac{f}{f + (1-f)e^{-\gamma\Delta c(\rho)}}, \quad \Delta c = \bar{c}_C - \bar{c}_H.$$

Appendix A separates exact Gaussian agreement probabilities from a deterministic mean-field surrogate obtained by applying softmax to expected counts. For positive gain, the surrogate gives the coalition more than its uniform share exactly when $\Delta c > 0$. For $b \neq 0$, this increases its squared bias; its MSE also contains honest and coalition variance. The Gaussian quantile $\sigma\Phi^{-1}(1/[2(1-f)])$ is an asymptotic reference for the median’s worst-case bias, so its intersection with $W_C|b|$ is a bias comparison, not a finite-sample error guarantee. With independent repetitions, in the long-window limit and on a correlation interval with $p_{CC} \geq p_{HC}$ throughout, a dependence strength $\lambda \geq \gamma(m-1)(N-1)/n$ makes the defended surrogate’s coalition weight non-increasing with correlation on that interval, where $N = n + m$. The realized softmax can have a different amplification boundary.

Numerically, with $n = 5$, $m = 2$, $\sigma_c = \sigma$, $h = 0.4\sigma$ (honest reporters agree with probability 0.22), $b = 4\sigma$ and $\gamma = 1$, credibility weighting is the best summing rule against *independent* liars (mean-squared error 0.87 against 1.45 for the mean and 0.57 for the median) and the worst rule against the *same* liars coordinating (1.84 at $\rho = 1$), while the median has similar rounded error at those endpoints (0.57): condition (i) has $\Delta c = -0.6$ at $\rho = 0$ and $+0.12$ at $\rho = 1$. The dependence-aware rule brings the coordinated case to 0.31, below the median, but pays for it at short windows and at low correlation (1.02 at $\rho = 0$), because the concentration statistic is estimated from few rounds. In the equal-variance, minority-coalition surrogate, independent biased reporters cannot amplify their weight: at $\rho = 0$, $\Delta c = (n-m)(p_{HC} - p_{HH}) \leq 0$. Unequal variances and finite-sample softmax fluctuations

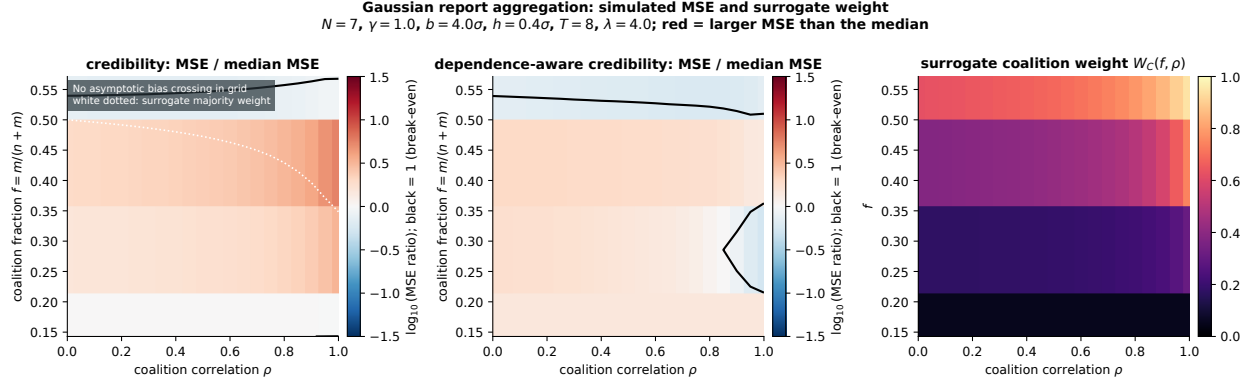


Figure 14. Phase diagram of the model ($N = 7$ reporters, $\gamma = 1$, $b = 4\sigma$, $h = 0.4\sigma$, $\sigma_c = \sigma$). Rows are discrete coalition sizes, $f = 1/7, 2/7, 3/7, 4/7$; contours interpolate the sampled grid. The row with $f \geq 1/2$ lies outside Appendix A's minority-coalition guarantee. Colour: $\log_{10}$ of the ratio of the estimator's error to the median's; black contour: break-even; white reference contours, where present, mark the surrogate's asymptotic bias-reference crossing and majority-weight boundary. The $N=7$ grid has no bias-reference crossing. These are analytical references, not exact boundaries for the realized estimator. Left: credibility weighting; middle: dependence-aware credibility (window of eight rounds, $\lambda = 4$); right: the mean-field coalition weight $W_C(f, \rho)$.

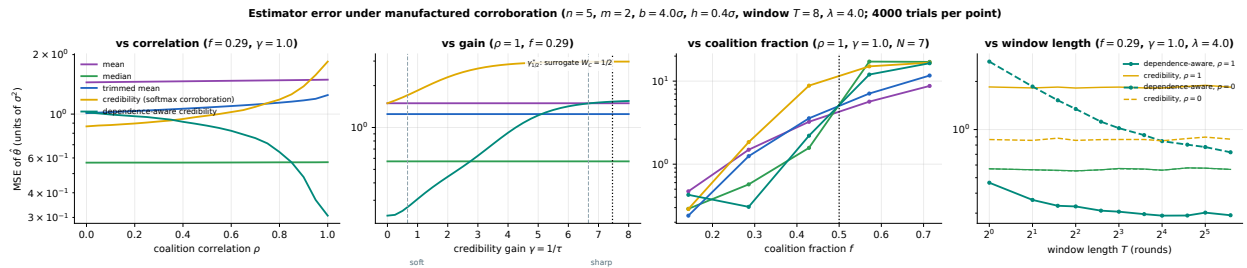


Figure 15. Estimator error (units of σ^2) against coalition correlation, credibility gain (the game's soft and sharp gains marked), coalition fraction, and the window length of the dependence penalty; 5+2 defaults with $\sigma_c = \sigma$, 4,000 trials per point. The coalition-fraction sweep includes values outside the minority-coalition guarantee.

require separate analysis. The median retains a uniform bound in coalition bias whenever $m < n$; this worst-case guarantee is lost at one half, which does not mean every estimator fails on every distribution.

Expected counts and expected weights can even predict different directions of amplification. For $n = 5$, $m = 2$, $\sigma = \sigma_c = 1$, $b = 20$, $h = 0.43$, $\rho = 1$, $\gamma = 1$, the surrogate gives $W_C = 0.295 > 2/7$, while 100,000 Monte Carlo trials give mean realized coalition weight 0.277, below the uniform share. The model supplies a mechanism and analytical references; the game’s realized weights must be checked directly.

Correspondence with the game. The scripted mechanism diagnostic directly measures lower average coalition weight than the uniform share in the tested alibi cells. This is compatible with the surrogate’s prediction when honest agreement is high, but does not validate a quantitative mapping from the Gaussian report model to symbolic testimony. The game also subtracts caught-lying penalties from credibility scores, so Table 5’s scores are not pure pairwise agreement counts. Neither the alibi-witness term nor the placement of the catch penalty is represented in the Gaussian model. The measured game weights, rather than the surrogate alone, establish the absence of average coalition-weight amplification in those cells. The initial 3+2 game has only 2+2 reporters after its incident victim is removed, so it lies outside the proposition’s strict-minority condition; the model’s n and m refer to actual reports, not initial players.

A sensor-fusion analogue. The rule was also applied to twelve sensors reporting a drifting scalar, with three compromised sensors (bias 3σ , $\rho = 1$, tolerance 0.4σ). Across 20 runs, credibility weighting assigns the coalition 0.254 ± 0.006 of total weight, only slightly above its uniform share of 0.25, compared with 0.099 ± 0.004 under independent compromised reports. Under coordination its RMSE is $0.99 \pm 0.03 \sigma$, versus 0.83 ± 0.02 for the mean and 0.58 ± 0.02 for the median. The dependence-aware variant reduces coalition weight to 0.044 and RMSE to $0.56 \pm 0.01 \sigma$. With independent compromised reports, the median remains best (0.58 ± 0.02), followed by credibility (0.60 ± 0.02), dependence-aware credibility (0.61 ± 0.02), and the mean (0.81 ± 0.01). These simulations illustrate a benefit of the dependence penalty under the chosen parameters; they do not establish a general repair guarantee for trust aggregation.

6.8 F7: a dependence-aware defense, and what a learned coalition does to it

The dependence-aware rule of Section 3.4 is the mechanism the model of Section 6.7 suggests: discount agreement that is concentrated in a small group. The archived comparisons below use the parameters in Appendix B; the corrected replication fixes those parameters before training. The archived study has three parts: against the scripted and the frozen learned coalitions, against a fresh coalition retrained with the defense in place, and inside the multi-generation loop.

Static coalitions. Against the scripted alibi the dependence-aware rule does what it was built for (Figure 16, Table 13): it keeps credibility’s honest-testimony behavior (false ejection

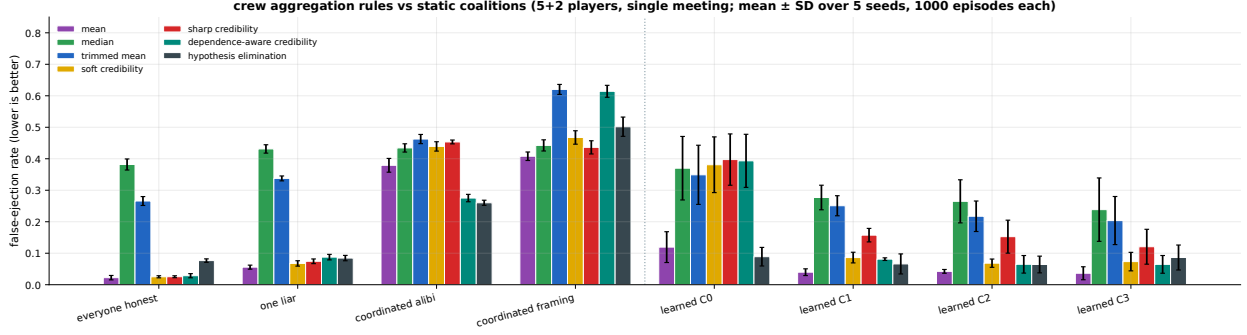


Figure 16. Archived measurements. False-ejection rates of the seven crew rules against the scripted coalitions and against the frozen learned coalitions C_0 to C_3 of the single-meeting multi-generation loop (5+2 players, spatial game; mean $\pm$ SD across five seeds, 1,000 episodes each).

0.029 against 0.025 for soft credibility) and reduces false ejections, $0.439 \rightarrow 0.275$, level with the hypothesis crew (0.260) and well below the mean (0.379). The strength response is monotone for the crew-rule placement (0.41, 0.36, 0.34, 0.27, 0.23, 0.22 at $\lambda = 0, 1, 2, 4, 6, 8$) with the honest rate flat at 0.02 to 0.03. Weighting ballots on its own is harmful: it raises the alibi rate to 0.47 at every strength up to 6 and at $\lambda = 8$ starts to eject honest players (0.14), because weighted ballots rarely tie and a meeting that plurality voting would leave undecided becomes an ejection. The rule is also harmful against the scripted framer ($0.47 \rightarrow 0.61$), whose signature, both members accusing and voting for the same innocent, is one that honest witnesses share whenever they converge on a suspect.

Frozen learned coalitions. The C0 coalition obtains false-ejection rates of 0.381, 0.397 and 0.393 against soft, sharp and dependence-aware credibility, respectively, versus 0.119 for the mean and 0.089 for hypothesis scoring. These differences are compatible with sensitivity to credibility weighting. They do not isolate a learned exploit of the catch-penalty placement, because the corresponding penalty intervention was performed on scripted coalitions, not these checkpoints. Frozen C0 can also produce different trajectories against different crews.

Later coalitions C1–C3, trained against learned crews, obtain lower false-ejection rates against the mean, credibility, dependence-aware and hypothesis rules (0.036 to 0.157), but remain more successful against median and trimmed mean (up to 0.277 and 0.251). The cross-play matrix thus shows substantial opponent specificity, rather than uniform robustness of every scripted rule.

Adaptive counterattack. A fresh coalition was trained for the C0 budget against soft credibility, against the dependence-aware crew rule, and against the crew rule combined with the dependence-weighted vote, and every trained coalition was evaluated against every defense (Figure 17, Table 14).

Table 13. Archived measurements. Alibi and framer refer to the coordinated scripts; other abbreviations follow Table 3. False-ejection rates of the seven rules in the spatial single-meeting game (5+2), mean across five seeds; across-seed SDs are 0.01 to 0.03 for the scripted rows and 0.01 to 0.10 for the learned ones.

Coalition	Mean	Median	Trim.	Soft cred.	Sharp cred.	Dep.-aware	Hyp.
Everyone honest	0.022	0.382	0.266	0.025	0.025	0.029	0.077
One liar	0.056	0.431	0.338	0.068	0.074	0.088	0.085
Alibi	0.379	0.434	0.463	0.439	0.453	0.275	0.260
Framer	0.408	0.442	0.620	0.468	0.436	0.614	0.502
Learned	0.119	0.370	0.349	0.381	0.397	0.393	0.089
C_0 (trained vs soft credibility)							
Learned	0.040	0.277	0.251	0.086	0.157	0.081	0.066
C_1							
Learned	0.042	0.265	0.217	0.068	0.153	0.065	0.064
C_2							
Learned	0.036	0.238	0.204	0.073	0.121	0.065	0.086
C_3							

Table 14. Archived measurements. Cross-evaluated false-ejection rates at 5+2 (mean $\pm$ SD across ten seeds, 1,000 episodes per cell). These coalitions, like C_0 in F3, create an incident in a minority of episodes (0.1 to 0.4 in replicates, Section 6.9), so most of the meetings scored here are incident-free.

Trained against	vs soft credibility	vs	
		dependence-aware rule	vs rule + weighted vote
Soft credibility (C_0)	0.418 ± 0.081	0.423 ± 0.071	0.536 ± 0.099
Dependence-aware rule	0.412 ± 0.090	0.420 ± 0.092	0.562 ± 0.086
Rule + weighted vote	0.384 ± 0.088	0.394 ± 0.083	0.558 ± 0.081

The defense that reduces the scripted alibi attack shows little benefit in these learned-coalition evaluations. The crew- rule placement has a small estimated effect (0.42 against 0.42 for the paper’s own C_0 , paired difference $+0.005 \pm 0.022$, $p = 0.508$; and a coalition trained against the rule is no weaker against soft credibility than one trained against soft credibility, 0.41 against 0.42), and the weighted vote helps the attacker: C_0 gains $+0.12 \pm 0.06$ against it without retraining ($p = 0.002$), and a coalition trained against it reaches 0.56; all three cells evaluated under the weighted vote exceed their corresponding unweighted cells. The archived training-log dependence summaries condition on agents surviving the ejection. Because ejection depends on the behavior being measured, those summaries cannot explain which pre-vote pairs the defense penalized. We therefore retain the outcome comparison but do not use those logged penalties to establish why it occurred.

Table 15 preserves the archived per-channel diagnostic for transparency. It was computed after ejection and excludes the ejected voter from pair rates and penalties. These quantities are conditional on survival and must be recomputed on pre-vote populations before they can support mechanistic comparisons of coalition and crew agreement.

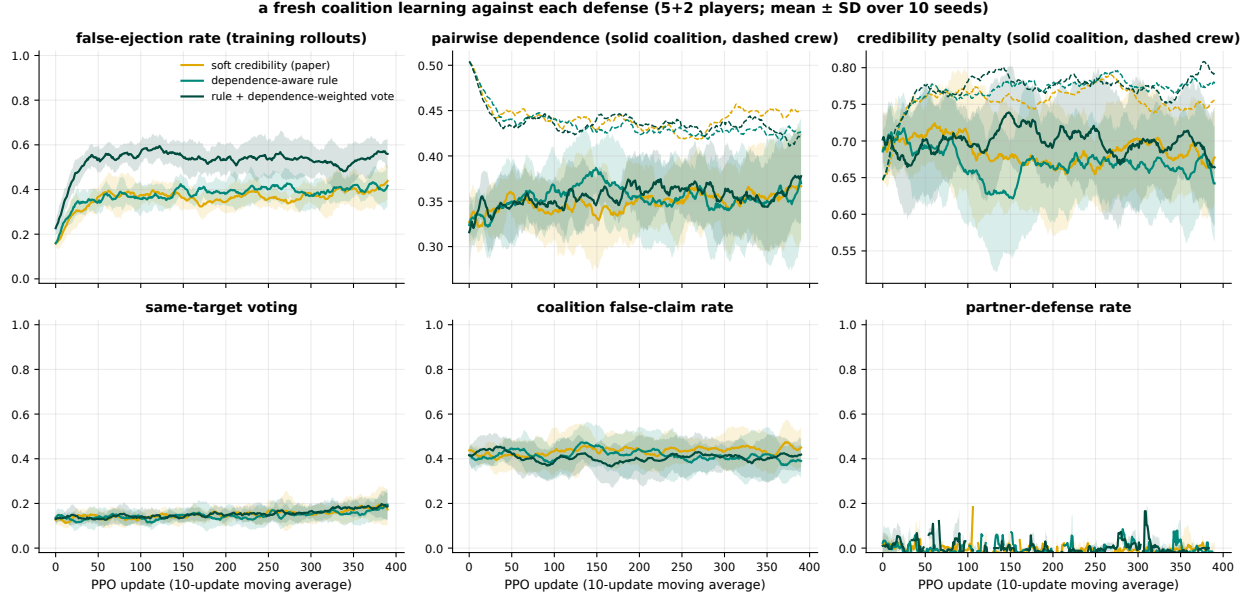


Figure 17. Archived coalition training against soft credibility, the dependence-aware rule, and rule + weighted vote (5+2; mean $\pm$ SD across ten seeds). Dependence/penalty panels, including the dashed crew lines, use post-ejection populations and are retained as archived diagnostics, not estimates of the weights applied to the full pre-vote electorate.

The archived channel table contains nonzero coalition vouching and support rates, so it cannot support the original assertion that learned coalitions never use these actions. Its survivor selection further prevents a reliable comparison with honest pairs. The observed cross-play still shows that success against the scripted alibi does not transfer to the tested learned coalitions; explaining that difference requires a corrected diagnostic and interventions on the suspected channels.

At 3+2, the coalition trained against the weighted tally obtains a false-ejection rate of 0.90 under that tally and 0.03 against soft credibility. Integer plurality can leave a two-versus-two tie, whereas unequal ballot weights can resolve it. This is a plausible mechanism, not proof that every such decision was won by the less-concentrated pair. The analogy to 50% contamination in the Gaussian model is qualitative; the voting game includes different information and rules.

Inside the multi-generation loop. With D_0 the dependence-aware crew and every later generation playing under the dependence-weighted vote (four generations, ten seeds, compared with the same generations and seeds of the undefended loop; Figure 18), the attacker’s stage-level point estimate is similar (coalition win 0.83 against 0.81 over the last three coalition stages, paired difference $+0.02 \pm 0.03$, $p = 0.053$; no equivalence test) and the defender stages are weaker (0.48 against 0.34 coalition wins averaged over the defender stages, $+0.14 \pm 0.12$, $p = 0.010$; terminal-meeting false ejection 0.39 against 0.24, $p = 0.002$). The archived coalition false-claim rate is lower with the defense (0.33 against 0.42), and games end sooner after defender stages (2.93 against 3.34 rounds). The logged dependence statistic has the

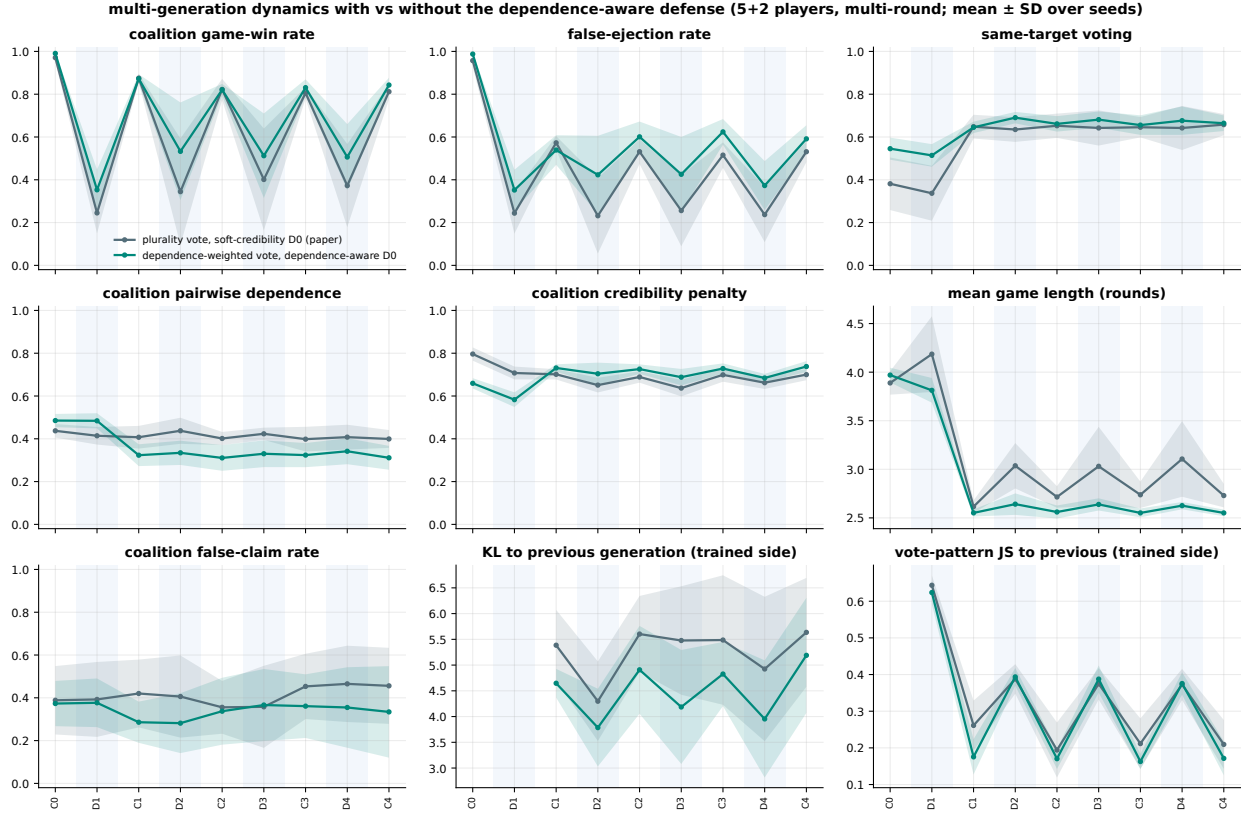


Figure 18. Archived measurements. The multi-generation loop with (teal) and without (grey) the dependence-aware defense over the first four generations, 5+2 players, multi-round game (mean $\pm$ SD across ten seeds). FE and communication rates summarize the final meeting; dependence panels retain the archived post-ejection selection and historical-window limitations described below.

Table 15. Archived post-ejection diagnostic, not valid as a pre-vote mechanism comparison. Per-channel agreement rates for surviving coalition pairs (C) and honest pairs (H), means across five seeds (the last row also averages over generations 1, 2, 3, 5, 7 and 10 of the single-meeting loop); “—” marks no recorded opportunities in this selected population. The joint-satisfiability channel is 0.87 to 1.00 for every coalition and 0.92 to 0.97 for honest pairs, and does not separate them.

Coalition	Vouching C /	Support C / H	Same vote C /	Penalty C / H
	H		H	
Scripted alibi	1.00 / 0.24	0.97 / 0.00	1.00 / 0.72	0.34 / 0.52
Scripted framer	0.00 / 0.22	0.00 / 0.00	1.00 / 0.87	0.75 / 0.58
Learned, trained vs soft credibility	0.07 / 0.19	0.08 / 0.00	0.27 / 0.47	0.72 / 0.74
Learned, trained vs dependence-aware rule	— / 0.22	0.08 / 0.00	0.25 / 0.39	0.73 / 0.75
Learned, trained vs rule + weighted vote	0.00 / 0.18	0.07 / 0.00	0.27 / 0.39	0.71 / 0.76
Learned C_1 to C_{10} of the loop (vs learned crews)	0.09 / 0.19	0.11 / 0.00	0.54 / 0.93	0.74 / 0.58

post-ejection selection problem described above; multi-round terminal diagnostics can additionally count the final meeting both as current and in history. Thus these numbers do not establish deliberate decorrelation or its causal role. The archived outcome comparison shows higher coalition win and terminal-meeting FE rates after defended defender stages, but this result needs corrected training and a diagnostic reconstructed at decision time.

6.9 F8: an adaptive attack on hypothesis elimination

The hypothesis crew is the strongest rule against the scripted alibi and holds the paper’s own C_0 to a false-ejection rate of 0.09 (Table 13). Both are evaluations against coalitions that were not trained against it. To ask whether its advantage is a property of the mechanism or of the mismatch between the mechanism and the attacks, a fresh coalition was trained for the C_0 budget directly against the hypothesis crew, alongside paired coalitions trained against soft credibility (the paper’s C_0) and against the mean, a strong summing baseline under the scripted alibi, and every coalition was evaluated against all three selected rules (Figure 19, Table 16; ten seeds at 5+2 and five at 7+2, the latter reported at the end of the section).

The advantage does not persist. A coalition trained against the hypothesis crew raises the crew’s false-ejection rate from 0.08 for the paper’s C_0 to 0.36, a 4.6-fold increase that holds in every seed (paired difference $+0.28 \pm 0.07$, exact sign-flip $p = 0.002$). Under adaptive attack the rule that was best against the scripted alibi at 5+2 has a similar point estimate

Table 16. Archived measurements. Cross-evaluated false-ejection rates at 5+2 in the spatial single-meeting game (mean $\pm$ SD across ten seeds, 1,000 episodes per cell). The soft-credibility row is a within-run replicate of the paper’s C_0 (Table 14) trained on a different machine; training is not bit-reproducible across machines, and the replicate’s rates lie within the seed spread of the original.

Trained against	vs soft credibility	vs mean	vs hypothesis elimination
Soft credibility (C_0)	0.369 ± 0.090	0.097 ± 0.050	0.079 ± 0.035
Mean	0.366 ± 0.068	0.100 ± 0.046	0.074 ± 0.030
Hypothesis elimination	0.091 ± 0.046	0.092 ± 0.037	0.359 ± 0.061

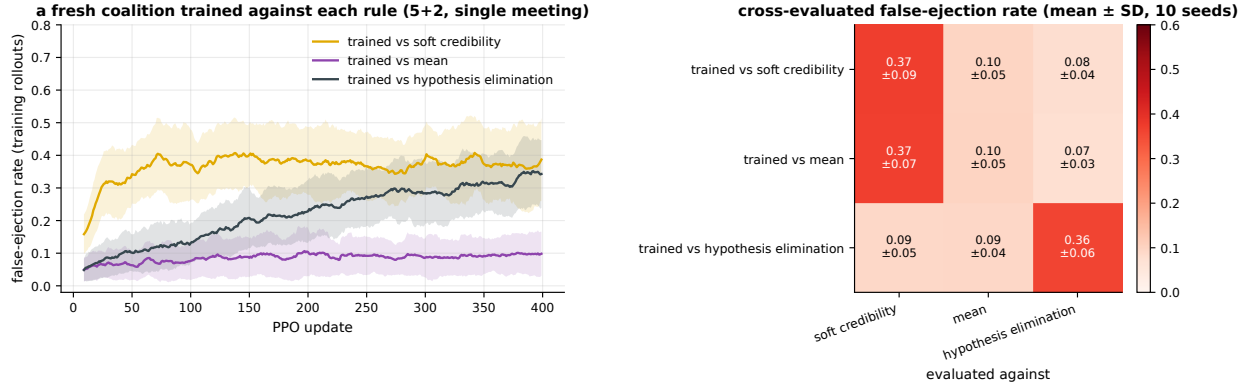


Figure 19. Archived measurements. Left: false-ejection rate on the training rollouts of a fresh coalition trained against soft credibility, the mean, and the hypothesis crew (5+2 players, mean $\pm$ SD across seeds, ten-update moving average). Right: the cross-evaluated false-ejection matrix of Table 16.

to soft credibility, which concedes 0.37 to its own adaptive coalition (hypothesis-minus-soft difference -0.010 ± 0.115 , $p = 0.787$; this is not an equivalence test), and about 3.6 times the rate of the mean, which concedes 0.10 to the coalition trained against it ($+0.26 \pm 0.08$, $p = 0.002$). The creator is ejected in 0.27 of the hypothesis-trained coalition’s incidents, where the archived scripted soft-credibility crew of F3 ejected C_0 ’s creator in 0.59. The training curve against the hypothesis crew is the slowest of the three and is still rising at the end of the budget (Figure 19), so a longer run is a relevant follow-up; an increasing noisy training curve is not a guaranteed lower bound on future performance.

Ballots and claim content. We reevaluated the saved checkpoints on 500 episodes per seed and corrected the diagnostic to include all agents that cast ballots, including the subsequently ejected player (Table 17). The original diagnostic excluded that player and evaluated hypothesis scores after the ejection, biasing both vote statistics and apparent confidence. The corrected calculation reconstructs the pre-ejection population.

The hypothesis-trained coalition uses fewer false claims and more no-information claims than coalitions trained against the summing rules. The hypothesis crew also casts many skips. These associations are consistent with an abstention-related vulnerability, but do not establish that the entire honest record contains nothing checkable or that inference is exact. For the hypothesis-trained coalition against its training defense, 0.65 of claims are no-information, 0.08 have the false label, same-target voting is 0.58, and crew skip rate is 0.66. Coalition ballots are necessary for the observed innocent’s unique plurality in 0.97 of false ejections. The mean top membership score is 0.52, not a uniformly flat distribution. A mean maximum membership score is a property of this simplified model, not a calibrated probability of error.

The ballot-removal diagnostic asks whether the same innocent would retain a unique plurality without coalition votes. A negative answer means coalition ballots were necessary for that particular ejection in the recorded tally. It does not imply that two coalition ballots alone were sufficient, that no crew member voted for the ejected innocent, or that crew-only voting would have produced no ejection at all. We do not make those stronger claims.

Table 17. Archived measurements. Corrected pre-ejection ballot decomposition at 5+2 (500 replayed episodes per seed, means across ten seeds). Each row identifies the coalition’s training defense and its evaluation defense; Hyp. denotes hypothesis scoring. “Incident” is the fraction of episodes with a real incident; “C votes needed” the fraction of false ejections in which the ejected agent would not have held the plurality without the coalition’s two ballots; “crew skips” the fraction of crew ballots that are skips; “same-target” the fraction of meetings in which the two coalition members vote for the same agent; “false claims” and “no-info claims” the fractions of coalition claims that are false or carry no information; “top membership score” the honest voters’ mean maximum normalized membership score under the hypothesis-elimination model, computed for every cell from the same record.

Trained vs → evaluated vs	Incident	FE	C votes needed	Crew skips	Same- target	False claims	No-info claims	Top membership score
Soft → Soft	0.23	0.37	0.51	0.64	0.21	0.43	0.08	0.72
Soft → Hyp.	0.23	0.09	0.85	0.38	0.20	0.42	0.08	0.72
Mean → Mean	0.20	0.10	0.41	0.10	0.16	0.37	0.14	0.73
Mean → Hyp.	0.19	0.07	0.89	0.37	0.17	0.37	0.14	0.73
Hyp. → Hyp.	0.90	0.37	0.97	0.66	0.58	0.08	0.65	0.52
Hyp. → Mean	0.91	0.08	0.33	0.06	0.46	0.08	0.65	0.52

Transfer. The hypothesis-trained coalition has lower false-ejection rates against the summing rules (0.091 against soft credibility and 0.092 against the mean) than against the hypothesis crew (0.359). Conversely, the soft- and mean-trained coalitions obtain 0.079 and 0.074 against the hypothesis crew while both obtain about 0.37 against soft credibility. Thus exploitation transfers between some attacks and defenses; it is not true that each rule is beaten only by its own trained coalition.

Among these direct attacks, the mean concedes 0.100 to the coalition trained against it, compared with 0.359 for hypothesis scoring and 0.369 for soft credibility. Mean aggregation does allow abstention and has nonzero skip rates in the diagnostic. Its lower observed harm therefore cannot be explained by a rule that never skips. Neither this comparison nor the rising training curve excludes a stronger attack with a longer budget or a different training procedure.

At 7+2 (five seeds), the hypothesis crew concedes 0.513 to its trained attacker versus 0.048 to the soft-trained coalition (difference $+0.465 \pm 0.048$, positive in all five seeds, exact sign-flip $p = 0.063$). Soft credibility concedes 0.374 and the mean 0.045 to their respective trained attackers. The hypothesis-trained coalition has a false-claim rate of 0.024 against its training defense and low transfer false-ejection rates against soft credibility (0.043) and the mean (0.037). These are archived-policy measurements and remain subject to the training-validity limitation of Section 4.1.

7 Interpretation

7.1 Scripted and adaptive evaluations

Within each tested configuration, a scripted evaluation measures a defense against the specified behavior. The F2 sweep separates aggregation rules under fixed testimony; the spatial F7 and F8 experiments test some of those rules against trained policies. The archived cross-play matrices show that the hypothesis-scoring crew’s advantage against scripted alibis does not persist against its own trained attacker. The mean has the lowest false-ejection rate under the direct adaptive attacks tested. This is a comparison at a finite training budget, not a guarantee that the mean is robust to every adaptive coalition.

The corrected F3 results establish a descriptive single-meeting comparison under repaired collection, independent evaluation and retained within-round private histories. They do not certify optimal responses or explain the gap by reasoning ability alone. The remaining archived learned-policy measurements are limited by the original rollout defect and other implementation changes. Their corrected reruns are needed to determine which multi-round and adaptive-defense patterns persist.

7.2 Distinguishing mechanisms

The scripted mechanism diagnostic measures reduced average coalition credibility weight under the alibi, and thus does not show average corroboration capture. Alibi testimony and the placement of caught-lying penalties affect outcomes in the tested interventions. Moving

that penalty outside a speaker’s weighted row improves several attacked cells, but does not uniformly dominate the mean. The Gaussian model formalizes a separate mean-field mechanism and cannot supply an exact threshold for this symbolic game.

For the hypothesis-scoring crew, uninformative claims, skip ballots and coordinated voting are consistent with an abstention-related vulnerability. The archived rule is not an exact Bayesian benchmark: it omits relevant constraints, smooths otherwise impossible hypotheses, and retains coalition claims in some branches. Its errors may arise from both an ambiguous record and misspecification. A causal test should vary the information content and abstention rule separately while holding other policies and episode draws fixed.

The original per-channel dependence diagnostic excluded ejected participants and, in multi-round terminal summaries, could count the current meeting twice. It cannot establish that learned coalitions coordinate less than honest crews or that decorrelation caused their success. The corrected ballot diagnostic includes every agent that actually voted, but a ballot-removal counterfactual still measures necessity for a particular ejection rather than proving that two coalition ballots alone were sufficient.

7.3 Training, information and response dynamics

Both learned teams use shared policies and team rewards. In the archived runs, the scripted crew had access to its full within-round observation history, whereas the non-recurrent actor saw a compressed last-sighting representation. Thus those comparisons combine policy learning with different retained information. Removing the explicit partner-identity feature also left role-dependent incident masks. The corrected replication retains private positional histories and removes the direct role-mask channel; its ablation still does not preclude inference from subsequent outcomes or test independently incentivized coalition formation.

The budget and reward controls retain positive last-mover gains in the archived runs, but they cover specific finite budgets and objectives. They do not establish equivalence, rule out training artifacts, or test the full lineage under the coordination ablations. Reward changes substantially affect defender-stage levels and the size of the last-mover gain.

F5 trains a fresh policy against a frozen opponent at every stage. Repeated outcome reversals and large distances between separately initialized policies do not demonstrate an attracting cycle, absence of a fixed point, or asymptotic non-convergence. Tests against policy populations, warm-started responses, restart controls and measured exploitability would be needed to address those questions. The observed last-mover gains are finite-horizon descriptive findings.

7.4 Whole-game outcomes

Multi-round play adds population changes, movement and repeated opportunities to create incidents. Game-win rate and rounds played therefore capture effects that a single-meeting metric cannot. The archived FE statistic, however, inspects only the terminal meeting. The separately recomputed cumulative incident-bearing FE probability has a different pattern. Future evaluations should report cumulative innocent ejections with and without incidents,

per-meeting rates, and game wins separately, rather than treating a lower terminal FE rate as reduced harm throughout the game.

8 Limitations

1. **Training validity.** The original asynchronous collector finalized unfinished fragments with zero terminal bootstrap and later credited rewards only to the remaining fragment. The collector is repaired and regression-tested. F1 and single-meeting F3 now report completed corrected replications; the remaining spatial learning and budget/reward/coordination results are provisional archives while their reruns continue. Completed results remain descriptive until the full planned analysis is available.
2. **Evaluation scope.** The corrected F1 study uses genuinely untrained and final checkpoints on independent held-out draws, with monitored curves on a separate stream. This replaces the archived reused-environment evaluation. Evaluation remains within the same benchmark distribution and does not measure transfer to unseen game rules or natural-language play.
3. **Information and ablation scope.** Archived learned actors did not retain the full private history used by scripted crews, and partner-feature ablation left role information in action masks. Both defects are repaired for the replication. C0-stage channel ablation still does not test a full response lineage or isolate all information and coordination mechanisms.
4. **Hypothesis model.** The archived rule used a uniform-room approximation and nonzero smoothing, and permitted incident victims in candidate coalitions. The corrected rule removes those features but remains a declared feasibility heuristic rather than an exact posterior under the game’s full data-generating process. Its new evaluations and adaptive attacks are pending.
5. **Diagnostic and outcome definitions.** The archived dependence summaries use post-ejection survivors and are not pre-vote mechanism estimates. The ballot diagnostic has been corrected separately. Archived multi-round FE is a final-meeting statistic and the old cumulative framing counter includes only rounds with real incidents. New records distinguish total harm, incident-bearing harm, and incident-free harm across all meetings.
6. **Statistical scope.** Seed counts are ten or five as specified in Section 5. The archived sign-flip tests are exploratory, unadjusted and require a sign-symmetry assumption. With five nonzero paired differences the minimum two-sided value is 0.0625. Null results do not establish equivalence, and multiple tables reuse seeds, policies and outcomes. Tables 5 and 6 now use ten independent evaluation replicates per population, but are a post-audit descriptive study. Common initial evidence does not make their response and vote transcripts identical across interventions.
7. **Sequential training.** Each stage is a finite-budget PPO-trained response from a fresh initialization, not a certified best response. Ten generations cannot establish asymptotic dynamics. Population-based and simultaneous training were not evaluated.
8. **Environment scope.** Four rooms, short horizons, symbolic claims and two imposers limit generalization to natural-language social deduction. Incidents require no

additional crew witness, but may occur with the partner present. Multi-round parity is checked after the meeting vote; a temporary parity after a kill does not itself terminate the game.

9. **Analytical scope.** The deterministic mean-field surrogate and asymptotic median-bias reference are distinct from finite-sample estimator risk. Dependence repair requires the stated agreement-ordering assumptions. The sensor simulation uses one parameter family.
10. **Reproducibility.** Local summary and episode artifacts permit substantial reaggregation, but many original checkpoints and full replays are absent. Archived results cannot all be reevaluated without retrieving missing artifacts or retraining. Some matched-seed runs used different machines; without a controlled comparison this remains a possible confound.

9 Conclusion

The completed corrected experiments show that aggregation rules and trained policies affect false ejections in this structured benchmark. Meeting-only coalition training increases held-out false ejections at all three population sizes. In the single-meeting spatial game, the learned defender ejects more innocents against C0 than the scripted crew, and coalition retraining raises this error at 5+2 and 7+2 but lowers its average at 3+2. The resulting policies transfer poorly to the scripted defender. These comparisons are descriptive and specific to the tested finite-budget training procedure.

The separate scripted replication finds a credibility disadvantage against coordinated alibis at the two larger sizes despite below-uniform average coalition weight. Alibi support and caught-lying penalty placement affect errors across conditions; their interventions do not provide a universally better rule. The analytical model supplies explicit conditions for weight amplification and attenuation in a deterministic surrogate, without calibrating an exact threshold for the symbolic game.

The audit corrected trajectory collection, evaluation, private-history retention, inference constraints and diagnostic populations. Multi-round response sequences, adaptive defense comparisons and controls still require their completed corrected measurements. The archived patterns cannot establish those findings for the repaired implementation or demonstrate asymptotic learning dynamics. The full replication and its prespecified analysis must be integrated before this working revision is ready for submission.

Acknowledgements

OpenAI Codex, using GPT-6, assisted with implementation review, correction and testing of research code, numerical reanalysis, and manuscript editing. Numerical measurements were produced by the executable experiments and analysis scripts described in this paper.

Data availability

The code, configurations and locally retained data are being assembled into a versioned reproducibility package alongside the corrected replication. A public archive identifier has not yet been assigned. Appendix C distinguishes archived artifacts from the corrected implementation and its new experiments.

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Appendix A. Proposition: corroboration capture under correlated testimony

Setting. Let $1 \leq m < n$, $N = n + m$, $f = m/N$, $\sigma, \sigma_c, h > 0$, $\gamma, \lambda \geq 0$, and $\rho \in [0, 1]$. Honest reports are $x_i = \theta + \varepsilon_i$ with independent $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. Coalition reports are $y_j = \theta + b + \eta_j$, with $\eta \sim \mathcal{N}(0, \sigma_c^2[(1 - \rho)I + \rho\mathbf{1}\mathbf{1}^\top])$, independent of the honest noise. Define $r = (x, y)$, $c_i = \#\{k \neq i : |r_i - r_k| \leq h\}$, and the realized estimator $\hat{\theta}_{\text{cred}} = \sum_i w_i r_i$, $w_i \propto e^{\gamma c_i}$. Dependence-aware weights additionally multiply by $e^{-\lambda \text{ex}_i}$, where $\text{ex}_i = \max_{k \neq i} D_{ik} - (N - 1)^{-1} \sum_{k \neq i} D_{ik}$ and D_{ik} is the agreement frequency across T independent repetitions with fixed model parameters.

Exact agreement probabilities. With $s^2 = \sigma^2 + \sigma_c^2$,

$$p_{HH} = \text{erf}(h/(2\sigma)), \quad p_{CC}(\rho) = \text{erf}\left(\frac{h}{2\sigma_c\sqrt{1-\rho}}\right), \quad p_{CC}(1) = 1,$$

$$p_{HC} = \Phi((h - b)/s) - \Phi((-h - b)/s).$$

Hence $\bar{c}_H = (n-1)p_{HH} + mp_{HC}$, $\bar{c}_C = (m-1)p_{CC} + np_{HC}$ and $\Delta c = \bar{c}_C - \bar{c}_H$ is non-decreasing in ρ .

Deterministic mean-field surrogate. Replace each random corroboration count by its group expectation. The resulting coalition weight and estimator are

$$W_C = \frac{f}{f + (1-f)e^{-\gamma\Delta c}}, \quad \tilde{\theta} = (1 - W_C)\bar{x} + W_C\bar{y}.$$

This substitution defines a surrogate. In general $W_C \neq \mathbb{E}[\sum_{j \in C} w_j]$, and it does not give an exact boundary for the realized estimator. Its exact bias and mean-squared error are

$$\text{bias}(\tilde{\theta}) = W_C b, \quad \text{MSE}(\tilde{\theta}) = W_C^2 b^2 + (1 - W_C)^2 \frac{\sigma^2}{n} + W_C^2 \sigma_c^2 \left(\rho + \frac{1-\rho}{m} \right).$$

The approximation $\text{MSE}(\tilde{\theta}) \approx W_C^2 b^2$ requires the displayed variance terms to be negligible relative to $W_C^2 b^2$.

Proposition (surrogate amplification and robustness).

1. *Weight amplification.* For $\gamma > 0$, $W_C > f$ if and only if $\Delta c > 0$, equivalently

$$(m-1)p_{CC} > (n-1)p_{HH} - (n-m)p_{HC}.$$

At $\gamma = 0$, $W_C = f$. When $\Delta c > 0$, W_C increases with γ and is non-decreasing in ρ ; $W_C > 1/2$ precisely when $\gamma\Delta c > \log((1-f)/f)$. Weight amplification increases the surrogate's squared bias for $b \neq 0$; a comparison of total MSE must also include the variance terms above. At $\rho = 1$ and negligible cross-group agreement, the amplification condition reduces to $(m-1) > (n-1)p_{HH}$. Negligible cross-group agreement requires separation relative to the noise scale, for example $(|b| - h)/s \gg 1$.

2. *Median robustness and the bias reference.* Because $m < n$, the pooled median M lies between the smallest and largest honest reports for every realization, irrespective of the coalition reports. Therefore

$$\text{MSE}(M) \leq \sigma^2 \mathbb{E} \left[\max_{1 \leq i \leq n} |Z_i|^2 \right] < \infty, \quad Z_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1),$$

uniformly in b and ρ . For odd N , writing $k = (N+1)/2$ and $Z_{(j)}$ for honest order statistics gives the sharper pathwise bound $x_{(k-m)} \leq M \leq x_{(k)}$, hence $|\text{bias}(M)| \leq \sigma \mathbb{E}[Z_{(k)}]$. As n, m grow with $m/N \rightarrow f < 1/2$, the latter upper bound tends to $B_\infty = \sigma \Phi^{-1}(1/[2(1-f)])$. Thus $W_C |b| = B_\infty$ defines an asymptotic bias-comparison reference, not a finite-sample MSE break-even boundary. For $0 < W^* = B_\infty/|b| < 1$, $W_C |b| > B_\infty$ is equivalent to

$$\gamma\Delta c > \log \frac{(1-f)W^*}{f(1-W^*)}.$$

When $\Delta c > 0$ and the right-hand side is positive, division by Δc gives the positive reference gain γ^* . Finite-sample error comparisons use the actual risks or the finite-sample

bound above. At $f = 1/2$, the median loses its uniform bound against arbitrarily displaced coalition reports; this does not imply that every estimator fails on every distribution.

3. *Dependence penalty.* In the long-window limit, for $m \geq 2$,

$$\begin{aligned}\bar{\text{ex}}_C &= \max(p_{CC}, p_{HC}) - \frac{(m-1)p_{CC} + np_{HC}}{N-1}, \\ \bar{\text{ex}}_H &= \max(p_{HH}, p_{HC}) - \frac{(n-1)p_{HH} + mp_{HC}}{N-1}.\end{aligned}$$

For $m = 1$, $\bar{\text{ex}}_C = 0$ and coalition correlation has no effect. The defended surrogate replaces $\gamma\Delta c$ by $\gamma\Delta c - \lambda(\bar{\text{ex}}_C - \bar{\text{ex}}_H)$. On any correlation interval where $m \geq 2$ and $p_{CC}(\rho) \geq p_{HC}$, its log-weight advantage depends on ρ through

$$\left[\gamma(m-1) - \lambda \frac{n}{N-1} \right] p_{CC}(\rho).$$

Consequently $\lambda \geq \gamma(m-1)(N-1)/n$ makes the defended surrogate weight non-increasing with correlation on that interval; a strict inequality makes it decrease where p_{CC} increases. For $5+2$, the threshold is 1.2γ . If $p_{CC} < p_{HC}$, the coefficient instead is $(m-1)[\gamma + \lambda/(N-1)]$, so the stated repair condition does not apply there. A comparison between $\rho = 0$ and $\rho = 1$ requires the first branch to hold throughout that interval.

Proof sketch. Groupwise softmax normalization gives W_C , and independence of the two groups gives the exact surrogate variance. The amplification and majority statements follow by rearranging the logistic expression. Fewer than half the reports are adversarial, so neither central rank can escape the honest range; for odd N at most m insertions can shift the median's honest rank, giving the stated order-statistic bounds. Gaussian symmetry gives the bias bound, and convergence of central order statistics gives its fixed-fraction limit. Finally, the strong law applied to each pair's agreement indicators gives the long-window matrix entries. Taking each row's actual maximum and collecting coefficients in p_{CC} yields the two dependence branches. $\square$

The tests check the Gaussian probabilities, limiting concentration statistic, simulation identities, and boundary cases. Monte Carlo experiments also exhibit finite-sample reversals of the surrogate's weight-amplification prediction, so they do not establish exactness of its realized-estimator thresholds.

Appendix B. Configurations, hyperparameters, and scripted policies

Table B1. Archived configurations and hyperparameters. The corrected hypothesis rule uses no pseudoweight; repaired collection, observation, evaluation and scoring procedures are specified in Sections 3–5 and Appendix C.

Setting	Value
Rooms	Hub, Electrical, Navigation, Medbay
Map, meeting-only game	star: Hub–Electrical, Hub–Navigation, Hub–Medbay
Map, spatial game	star plus Electrical–Navigation
Free-play turns	6 (meeting-only), 8 (spatial)
Evidence, meeting-only game	random walk on the map, stay probability 0.35, one incident placed by the episode process
Evidence, spatial game	movement actions; two tasks per agent; incident requires one crew target and no other living crew in the room; the partner may be present; automatic report at the end of free play
Incident time	announced publicly by the report, so the scene window is the incident time step
Meeting	claim round (structured claim with polarity and confidence), response round, plurality vote with skip; ties do not eject; roles not revealed on ejection; random speaking order
Coalition	two impostors; partner identity in the observation; shared policy; shared reward; learned private symbol channel with four symbols (spatial game only)
Actor and critic	two-layer MLPs, 128 hidden units, non-recurrent
PPO	learning rate 3×10^{-4} , clip 0.2, $\gamma = 1$, GAE $\lambda = 0.95$, up to 8 epochs (target KL 0.05), 4 minibatches stratified by phase, value coefficient 0.5, entropy coefficient 0.02
Updates per stage	annealed to 0.005, gradient clip 0.5
	400 standard; 800 or 1,600 for defender-budget controls
Generations (multi-generation loops)	10 (5+2 and 3+2), 3 (7+2), 4 (defended loop, single-meeting reward control), 2 (multi-round reward control)

Setting	Value
Rollout per update	64 total episodes across 64 environments (F1); target of at least 32 completions across 32 environments (archived spatial training)
Evaluation	400 episodes every 20 updates (F1); 1,000 cross-play episodes per stage (F3, F4, controls); 1,000 per stage and 500 per cross-play cell (multi-generation)
Seeds	10 (F1–F4; F8 at 5+2; 5+2 multi-generation loops, defended loop and counterattack), 10 (3+2 loop), 5 (7+2 loop), 5 (F8 at 7+2, static learned-policy evaluations, controls); see Section 5 for evaluation-seed conventions
Credibility temperatures	1.5 (soft), 0.15 (sharp)
Dependence-aware defense	$\lambda = 4$, window of three meetings, top-1 partner
Hypothesis crew	honesty probability 1/2, ejection threshold 1/2, excluded-hypothesis pseudoweight 10^{-3}

Suspicion weights of the scripted crew. Own evidence: an agent the voter saw in the incident room during the scene window +3.0; an agent the voter saw elsewhere −3.5; an agent the voter could not place +1.0; a speaker with no informative claim +0.4. Testimony: a claim the voter personally knows to be false +4.0 on its speaker (the caught-lying penalty); a publicly self-contradictory claim +3.0 on its speaker; a claim placing an agent in the incident room during the window +1.5; a claim by another speaker placing an agent elsewhere during the window −2.0 (the alibi-witness term); a speaker’s own claim placing itself elsewhere −0.5; a positive was-with claim −1.5 on the subject and −0.5 on the speaker; an accusation +0.8 on its target, a contradiction +0.4. The vote threshold is 0.5. Credibility is −3.0 when the voter catches the speaker lying; otherwise −2.0 for a self-contradictory claim set; otherwise the count of other informative speakers whose claims are jointly satisfiable. The caught and contradictory branches replace the corroboration count.

Scripted coalitions. The full behavioral definitions are in Section 3.3. In particular, alibi support is asymmetric (creator self-location plus partner sighting/defense), and the framer makes scene-placement claims rather than necessarily truthful testimony. The archived scripts’ spatial movement inspected adjacent-room occupancy; the corrected scripts use their own sightings and other legitimately available information.

Appendix C. Reproduction

The public source package contains corrected scientific code in `code/`, manuscript sources and figures in `paper/`, and the publication audit record in `cahnges.md`. See `code/README.md` for reproduction commands. `paper/build.sh` regenerates PDF and TeX. Historical launchers,

live status, raw data and checkpoints are excluded.

Fresh runs use the same experimental design with a new packaging fingerprint and protocols. The original 240-job protocol is `code/provenance/original_study_protocol.json`; existing corrected measurements retain their original source and run provenance. Passing tests cannot validate old-collector results. Reruns need a fresh results directory, code version, complete checkpoints, environment configuration and independent evaluation seeds.

Post-audit replication protocol. The correction package specifies 240 jobs containing 1,255 learner stages in `code/experiments/submission/run_study.py`. It preserves the original seed counts, 400-update standard budget, 800/1,600-update defender controls, 64 completed episodes per F1 update, and 32 completed episodes per spatial update. A frozen source manifest and job-specific completion records distinguish these runs from the archives. The primary final evaluation stream starts at seed 1987654321; monitored training curves use the separate 987654321 stream. All evaluations finish fixed episode-index batches, and evaluation policies have isolated sampling generators. F1 retains an untrained checkpoint and checkpoints every 20 updates; spatial runs retain checkpoints every 50 updates. Completed multigeneration evaluations retain per-game and per-meeting CSVs.

The revised spatial observation adds within-round private histories, and the partner-information intervention removes direct identity, legality-mask and message-missingness channels. Scripted movement no longer inspects unseen adjacent rooms. Hypothesis scoring applies hard public-role and joint positional constraints without pseudoweights for impossible hypotheses; it remains a feasibility heuristic, not an exact posterior for the full game. Dependence is computed over the complete electorate with each meeting counted once. New harm metrics include any innocent ejection and total innocent ejections across all meetings, including incident-free meetings. These changes define the corrected replication. Tables 2 and 7 and Figures 1 and 3 report completed F1/F3 groups; Tables 5 and 6 report the completed mechanism supplement. Other empirical tables and figures remain archived until their replacements are checked. Detailed changes and evidence status appear in `cahnages.md`; `code/README.md` documents reproduction using `code/submission_protocol.json`.

The supplemental mechanism study uses ten independent evaluation replicates per population, 500 episodes per rule/condition/intervention cell, four scripted conditions, three summing rules and three mechanism variants (540,000 episodes in total). Initial evidence is matched across the nine rule/variant cells within each condition and replicate. All episode outcomes and the baseline soft-rule voter records are retained. Both response actions and votes use the selected variant; these are full mechanism interventions. The portable supplement scripts and fresh-run protocols are under `code/audit/submission/`; existing measurements retain their original frozen protocol. This study is post-audit and descriptive. The checkpoint-dependent F8 ballot replay has a separate fixed protocol and awaits its newly trained input policies; the archived Table 17 is not a measurement of the repaired hypothesis rule.